

Urinary Tract Infection in Adults

Angela Huttner | Julia A. Bielicki | Fadi Haidar Sr.

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KEY POINTS

- Urinary tract infection (UTI) is a common problem for all ages, affecting primarily healthy females and individuals with underlying genitourinary abnormalities. With advancing age, healthy males become increasingly affected.
- The optimal approach to UTI management requires evaluation of the clinical presentation and critical interpretation of urine culture results, together with an assessment of underlying comorbidities and urologic abnormalities.
- The evolution of antimicrobial resistance in common uropathogens, including *Escherichia coli*, requires continuing reassessment of empiric antimicrobial treatment.
- Asymptomatic bacteriuria is common in many populations, but treatment is indicated only for individuals who undergo invasive urologic procedures associated with mucosal bleeding. There is ongoing debate as to whether pregnant women with asymptomatic bacteriuria universally require antibiotic therapy.
- Indwelling devices in the urinary tract become coated with biofilm following insertion, and the presence of the biofilm will have an impact on the urine culture and outcomes with antimicrobial therapy.
- Small renal abscesses (<5 cm) can usually be managed medically with prolonged courses of antimicrobial therapy, but larger abscesses including most perinephric abscesses require drainage in addition to antimicrobial therapy.
- Genitourinary tuberculosis is one of the most common forms of extrapulmonary tuberculosis and usually presents as reactivation with involvement of only one kidney or the ureter.
- *Schistosoma haematobium*, an endemic parasite in many regions of the world, localizes to the bladder wall; acquisition of infection with this parasite is a risk for local populations and travelers to these areas.

Urinary tract infection (UTI) of the bladder, kidney, or prostate is one of the most common human infections. Infecting organisms are usually bacteria; fungi, viruses, or parasites are less frequent pathogens. Other manifestations of genitourinary tract infection are renal and perinephric abscesses, emphysematous cystitis and pyelonephritis, xanthogranulomatous pyelonephritis, and pyocystitis. Disseminated viral infections (e.g., mumps, cytomegalovirus, and other herpes viruses) and noncandidal fungal infections

(e.g., blastomycosis and histoplasmosis) may also involve the urinary tract but are not discussed in this chapter.

DEFINITIONS

UTI results from the presence of bacteria or other microorganisms in the urine or genitourinary tissues. The term “bacteriuria” describes isolation of any bacteria in the

urine, although in practice it usually refers to isolation of organisms in concentrations that meet specific quantitative criteria. Infection is asymptomatic when the urine culture result meets quantitative criteria for bacteriuria without signs or symptoms attributable to infection. Symptomatic UTI may manifest as bladder infection (cystitis or lower-tract infection), kidney infection (pyelonephritis or upper-tract infection), or prostate infection (acute or chronic bacterial prostatitis). Acute UTI occurring in women usually manifests as cystitis.¹⁻³ Pyelonephritis is much less frequent.²

UTI commonly recurs. It is useful to distinguish between “reinfection” and “relapse.” Reinfection is a new infection following eradication of a prior infection and typically occurs with a different organism. “Relapse” occurs when the initially infecting organism persists in the urinary tract despite antimicrobial therapy. Eradication was never achieved. In the case of relapse, the same organism is isolated. Recurrent UTI is arbitrarily defined as a relapse if the recurrence presents within 2 weeks of completion of therapy for the initial infection and the isolated uropathogen is the same. An infection that occurs more than 2 weeks after completion of therapy (with full resolution of symptoms in the interval) is considered a reinfection even if the same organism is isolated—which may be the case in up to 30% of *E. coli* reinfections. This finding is assumed to be a consequence of failure of the antimicrobial therapy to eliminate virulent strains from gut or vaginal reservoirs.² Intracellular persistence of *E. coli* in uroepithelial cells is an alternative mechanism proposed to explain same-strain recurrence, as observed in animal studies and in children with recurrent infection.⁴⁻⁶ However, prospective studies in women document periurethral colonization before bladder infection onset for most episodes,⁷ and the contribution of an intracellular reservoir to recurrent acute uncomplicated cystitis in humans remains uncertain.

Historically, UTI has been divided into uncomplicated and complicated infection,^{8,9} yet full consensus on the definition of “complicated UTI” was elusive. It was agreed that host (and not pathogen) factors complicated infection, but exactly which factors and their relative importance were debated. In broad strokes, they included obstruction (pelvic/urethral junction obstruction, ureteric or urethral strictures, prostate hypertrophy, urolithiasis, tumor); neurologic impairment; the presence of urologic devices (indwelling catheters, stents); functional or structural abnormalities (vesicoureteral reflux, ileal conduit); metabolic/congenital diseases (nephrocalcinosis, polycystic kidneys); and immunologic impairment. The concept was counterintuitive for some, as severe pyelonephritis with septicemia in a patient without these baseline factors was considered uncomplicated UTI.

The paradigm is changing. The Infectious Disease Society of America (IDSA) proposes that “uncomplicated UTI” be defined as a clinical syndrome confined to the bladder, characterized by local bladder signs and symptoms such as dysuria, urgency, frequency, and suprapubic pain. Signs or symptoms suggesting systemic illness (fever, flank pain, costovertebral angle tenderness) are lacking, and the definition does not require pyuria or a documented uropathogen. By this paradigm, uncomplicated UTI can occur in patients with underlying urologic abnormalities and immunocompromise, and recurrent UTI can be

uncomplicated. The presence of an indwelling catheter, however, is still considered to be a complicating factor.

“Complicated UTI” is now defined as a clinical syndrome including signs and symptoms suggesting infection extending beyond the bladder, including fever ($\geq 37.8^{\circ}\text{C}$), chills or rigors, unstable vital signs, significant fatigue or malaise beyond baseline, flank pain, or costovertebral angle tenderness. It thus includes all pyelonephritis. This definition also drops the requirement for pyuria and/or a documented microbial pathogen, but their absence suggests a different diagnosis unless extenuating circumstances (neutropenia, prior receipt of antibiotics) are present (IDSA guidelines update 2024, personal communication).

GENERAL CONCEPTS

HOST DEFENSES OF THE NORMAL URINARY TRACT

In the era of culture-based microbiologic methods, the genitourinary tract was thought to be sterile apart from the distal urethra. The normal flora of the distal urethra plays an important role in host defense by preventing colonization at this site by potential uropathogens. The flora includes aerobic bacteria that are common skin commensals, such as coagulase-negative staphylococci, viridans group streptococci, and *Corynebacterium* species.^{10,11} There is also a large and complex anaerobic flora.¹¹

Yet recent studies using 16S ribosomal RNA sequence analysis of bladder urine collected by suprapubic aspiration or urethral catheterization also describe a urinary microbiota of predominantly nonpathogenic gram-positive organisms, with some other species present.^{12,13} This microbiota shows substantial heterogeneity among individuals and variation with age.¹⁴ The role of the microbiota has been suggested to be protective by means of biofilm barrier, nutrient competition, or priming of host defenses.

Urine is a good nutrient source for most bacterial species. The most important host defense that maintains sterility of the urine is normal, unobstructed voiding. A complex array of urine and uroepithelial-cell components also contributes to maintenance of nonpathogenic urine in the normal genitourinary tract (Table 38.1).¹⁵⁻¹⁷ Inhibitors of bacterial adherence to uroepithelial cells prevent bacterial persistence once they have entered the urinary tract.¹⁶ For example, Tamm-Horsfall protein prevents attachment of *Escherichia coli* by binding to its type 1 fimbria adhesin (FimH).¹⁸ It also may have an immunomodulatory role through activation of the innate immune response by a Toll-like receptor (TLR)-4-dependent mechanism.¹⁶

Bacterial adherence is also prevented by the surface mucopolysaccharide–glycosaminoglycan layer of the uroepithelium, by urine immunoglobulin G (IgG) and secretory immunoglobulin A (IgA), and by some low-molecular-weight oligosaccharides present in the urine. The relative *in vivo* importance of any of these is not yet established. Despite the many components contributing to maintaining sterility of the urine, bacteriuria is readily established once normal voiding is impaired. In the complicated urinary tract, infection occurs through increased entry of organisms into the bladder or kidney, which may be attributed to the use of urologic devices,

Table 38.1 Some Host Defenses Other Than Voiding that Contribute to Maintaining Sterility of Urine

Defense	Example(s)
Normal flora	Various microbiota of the genitourinary tract (urethra, bladder, vagina, etc.)
Urine characteristics	pH, osmolality, concentrations of organic acids
Urine proteins	Tamm–Horsfall protein (uromodulin), secretory immunoglobulins, lactoferrin, lipocalin, cationic peptides (defensins, cathelicidins), siderophores
Innate and cellular immune response	Toll-like receptors, polymorphonuclear leukocytes, chemokines/cytokine production, antimicrobial peptides, myeloid dendritic cells, macrophages, mast cells
Uroepithelium	Mucopolysaccharide layer, glycoprotein plaque (uroplakins), epithelial barrier, shedding of superficial epithelial cell layer
Prostate secretions	Chemokines, immunoglobulins

turbulent urine flow, or ureteric reflux. Organisms may then persist despite other host defenses when infected urine is retained if voiding is incomplete, or in biofilm on urologic devices.

IMMUNE AND INFLAMMATORY RESPONSES TO URINARY TRACT INFECTION

The innate immune system provides the major immunologic response to UTI.¹⁷ The intensity of response is determined by the interactions of microbial pathogenicity, individual genetic regulation, and site of infection.^{19,20} Unique *E. coli* strains have a variable capacity to stimulate or evade activation of the innate response. Uropathogenic strains that cause symptomatic infection induce a strong innate response, whereas strains isolated from asymptomatic bacteriuria evoke a limited response.^{21,22} Strains successfully evading immune activation may have a pathogenetic advantage for establishing bladder colonization and persistent infection.^{23,24} Host genetic polymorphisms affecting the innate response predispose to acute pyelonephritis or asymptomatic bacteriuria.^{20,25}

Infecting organisms that gain access to the bladder adhere to uroepithelial cells, and stimulation by bacterial lipopolysaccharide leads to activation of these cells.²⁶ Recognition is through TLRs (TLR-4, TLR-2, TLR-5, and TLR-11) and activation promotes cytokine production, particularly interleukin (IL)-1, IL-6, and IL-8 (chemokine [C–X–C motif] ligand 1 [CXCL-1]). These cytokines recruit neutrophils and other immunocompetent cells to the kidney and bladder.^{16,17,20} The chemotactic cytokine IL-8 is released at the mucosal site and induces a rapid influx of neutrophils into the bladder, with subsequent phagocytosis and clearance of bacteria. Macrophages and mast cells residing in the submucosa are also recruited. Organisms

may also infect superficial bladder epithelial cells, with subsequent shedding of these cells. This innate response rapidly clears most uropathogenic *E. coli* organisms from the bladder, but it does not produce a sterilizing immunity in murine models.²⁷ In humans, bacteriuria often persists despite marked pyuria.

Urine and serum IL-6 concentrations correlate with severity of infection. The highest levels occur in patients with pyelonephritis. Systemic elaboration of IL-1 β and IL-6 produces fever and activation of the acute-phase response. The acute inflammatory infiltrate of polymorphonuclear leukocytes that develops in renal tissue during pyelonephritis limits bacterial spread and persistence within the kidney but also contributes to tissue damage and renal scarring.

IgA-producing plasma cells are found in higher numbers in the bladder submucosa of patients with bacterial cystitis than in healthy controls. However, acute cystitis is associated with a reduced or undetectable serologic response, presumably reflecting the superficial nature of the infection.¹⁷ The local immune response is of short duration and is reactivated for each infection. This limited response to bladder infection may explain why recurrence with the same *E. coli* strain is observed in some women with acute cystitis. However, animal studies have reported some protection against same-strain reinfection mediated by systemic and local antibodies.²⁷

A vigorous local and systemic humoral immune response occurs in patients with pyelonephritis.^{15,21} The antibody response is directed against surface antigens of the infecting bacteria, including O antigens, and surface proteins such as the type 1 (FimH) and P fimbriae, both major adhesins of *E. coli*.^{27,28} IgM antibodies dominate in the first episode of upper UTI, but subsequent episodes are characterized by an IgG response. In pyelonephritis, elevations of IgG antibodies to lipid A correlate with severity of renal infection and parenchymal destruction. There is also a substantial urinary IgG and secretory IgA antibody response. Nonetheless, the protective role, if any, of the antibody response in pyelonephritis is not clear. Bacteria often persist in the renal parenchyma despite high levels of specific antibodies. In addition, the frequency of UTI is not increased in women who do not produce secretory IgA.

Cell-mediated immunity appears to have a limited role in the host defense against UTI. A small number of mucosal T-lymphocytes are present throughout the urinary tract, and both CD4⁺ and CD8⁺ T-cells can be found in the submucosa and lamina propria of the bladder and urethra. Recruitment of B- and T-lymphocytes to the bladder wall is observed with secondary infections. T-cell-derived proinflammatory cytokines also stimulate renal tubular epithelial cells to produce IL-6, which may increase IgA secretion of committed B cells.²⁷ However, women infected with human immunodeficiency virus (HIV) and very low CD4⁺ counts do not have increased susceptibility to or severity of UTI,²⁹ suggesting that cell-mediated immunity is not an essential defense.

URINE CULTURE

The definitive diagnosis and appropriate management of UTI usually require microbiologic confirmation by urine culture. Urine specimens for culture should always be

obtained before antimicrobial therapy is initiated because urinary excretion of antimicrobial agents rapidly sterilizes urine. Once collected, the specimen should be forwarded promptly to the laboratory. Organisms present in small quantitative counts (i.e., contaminants) grow readily in urine at room temperature and reach high quantitative counts within a few hours. If the specimen is delayed in reaching the laboratory, it should be refrigerated at 4°C until transported.

A urine specimen for culture must be collected with a method that minimizes contamination. A clean-catch voided specimen without additional periurethral cleaning is usually appropriate. When patients cannot cooperate for the collection of a voided specimen, urine may be collected by an in-and-out catheter. For men, a specimen may be obtained in an external condom catheter after application of a clean condom catheter and collecting bag.³⁰ Urine samples may also be collected by suprapubic aspiration or directly from the renal pelvis when percutaneous drainage of an obstructed urinary tract is necessary. Specimens obtained from patients with short-term indwelling catheters should be collected by puncture of the catheter port. For a long-term indwelling catheter, two to five organisms are present in the catheter biofilm at any time, so urine collected through the catheter will be contaminated by organisms present in the biofilm.³¹ The long-term indwelling catheter should be removed and replaced by a new catheter, and a specimen of bladder urine obtained through the newly placed catheter.^{30,31}

The standard quantitative criterion for diagnosis of UTI with voided specimens was traditionally considered to be an organism count of $\geq 10^5$ colony-forming units (CFU)/mL of a potential uropathogen. This cutoff was set in an era when urine was routinely collected in the morning from patients in a fasting state and thus was highly concentrated. More recently, the cutoff has been considered to be $\geq 10^4$ CFU/mL, but recent evidence suggests that even this threshold is likely too high—particularly given that patients are increasingly aware of the importance of augmented hydration and may thus present in a setting of high fluid intake—and that thresholds in general may need to be pathogen specific. For example, *E. coli* at amounts of 10^2 CFU/mL, or even in lower amounts detectable only by polymerase chain reaction (PCR), appear sufficient to induce clinical symptoms.³²

Indeed, while application of a quantitative standard may be useful for the diagnosis of asymptomatic bacteriuria, for symptomatic patients, quantitative culture results must be interpreted in the context of the clinical presentation and the method of specimen collection (Table 38.2). Bacteria require several hours of incubation in bladder urine to achieve concentrations $\geq 10^4$ CFU/mL; some patients with frequency or diuresis may not retain urine for the time needed. Quantitative counts may also be lower when infection is caused by fastidious organisms, the patient is receiving a urinary antiseptic, or is substantially self-hydrating to treat the infection. For symptomatic men, a single urine specimen in which $\geq 10^3$ CFU/mL of a uropathogen is isolated is considered diagnostic for bladder bacteriuria on the basis of paired comparisons of voided specimens and suprapubic aspirates.

A urine specimen obtained by suprapubic aspiration or other percutaneous collection method such as renal

Table 38.2 Typical Quantitative Counts of Bacteria in the Urine for Microbiologic Diagnosis of Urinary Tract Infection in Patients not Receiving Antimicrobial Therapy

Collection Method	Quantitative Criteria (CFU/mL)
Voided Specimen	
Asymptomatic women or men	$\geq 10^{5a}$
Women: Cystitis	$\geq 10^{3b}$
Women: Pyelonephritis	$\geq 10^{4c}$
Men: Symptomatic	$\geq 10^3$
Men: External condom collection	$\geq 10^5$
Catheter	
In-and-out indwelling ^d	$\geq 10^2$
Asymptomatic	$\geq 10^5$
Symptomatic	$\geq 10^2$
Suprapubic or percutaneous aspiration	Any growth

^aTwo consecutive specimens are recommended for women.

^bFor *Escherichia coli*, values of $\geq 10^2$ colony-forming units (CFU) per milliliter are likely clinically significant.³²

^cIn 95% of cases, values $\geq 10^5$ CFU/mL.

^dA long-term catheter should be replaced and the specimen

pelvis drainage is assumed to be a clean specimen, and any quantitative count of an organism should represent true bacteriuria. However, in specimens collected by an in-and-out catheter, contaminating organisms are introduced from the periurethral area and a quantitative criterion of $\geq 10^2$ CFU/mL is recommended. Other relevant considerations in interpreting a urine culture result include the number and type of organisms isolated. A single infecting organism is usual, but in patients with complicated UTI, particularly those with indwelling urinary devices, more than one organism is frequently present. Commensal bacteria of the urinary microbiota, such as *Lactobacillus* spp., and normal skin flora such as coagulase-negative staphylococci, should usually be considered contaminants when they are isolated from voided urine specimens. In premenopausal healthy women, group B streptococci and *Enterococcus* species isolated in any quantitative count are also usually contaminants.^{33,34}

PHARMACOKINETIC AND PHARMACODYNAMIC CONSIDERATIONS FOR TREATMENT

Therapeutic success in the treatment of cystitis and pyelonephritis depend on antimicrobial levels in the urine and renal tissue, respectively; renal-tissue concentrations correlate with serum concentrations.³³ Urine concentrations are determined by glomerular filtration, active tubular secretion, and tubular reabsorption, all influenced by urine pH, protein binding, and the medicine's molecular structure; Table 38.3 describes urinary excretion of various antimicrobials. Cystitis and pyelonephritis may be successfully treated with antimicrobial agents at minimum

inhibitory concentrations (MICs) to which the infecting organism would not usually be considered susceptible. The “intermediate” susceptibility designation reported by the laboratory implies clinical efficacy in body sites where antimicrobial agents are physiologically concentrated, such as the urine. Thus when an organism isolated from the urine is reported to have intermediate susceptibility to an antimicrobial agent, the drug is usually appropriate for treatment of UTI with that organism.

Some agents' bactericidal activity is modified by urine pH. Penicillins, tetracyclines, and nitrofurantoin are more active in acidic urine, and aminoglycosides, fluoroquinolones, and erythromycin are more active in alkaline urine. This variability has not, however, been shown to be relevant for therapeutic outcomes, with the exception of methenamine salts, for which an acidic pH is necessary to release formaldehyde, the active component.

The prostate is a unique compartment for consideration of antimicrobial efficacy. There are no active antibiotic transport mechanisms for the gland, and most antibiotics penetrate poorly into prostate tissue and fluid.³⁵ The interior of the gland is an acidic environment. Drug entry and activity are determined by concentration gradient, protein binding, lipid solubility, molecular size, local pH, and the strength of acidity (pK_a) of the antimicrobial agent. Alkaline drugs such as trimethoprim (TMP) diffuse into the prostate and are trapped, with high concentrations achieved, but the drug remains in an inactive, ionized form. Fluoroquinolones and macrolides, however, penetrate well and remain active.

Current pharmacodynamic models for antimicrobial treatment of infection distinguish between time-dependent and concentration-dependent bacterial killing. Killing by β -lactam agents is time dependent, as the therapeutic efficacy depends on how long the antibiotic's concentration remains above the MIC of the infecting organism. Bacterial killing by fluoroquinolones and aminoglycosides is more concentration dependent; therapeutic efficacy is measured by the ratio of peak antimicrobial concentration to MIC, or the ratio of the area under the curve (AUC) to MIC. In contrast, nitrofurantoin's activity appears to be pathogen specific, with time-dependent killing of *E. coli* yet concentration-dependent killing of *Enterobacter cloacae* observed.³⁶

USUAL PRESENTATIONS OF URINARY TRACT INFECTION

ACUTE UNCOMPLICATED URINARY TRACT INFECTION (CYSTITIS)

EPIDEMIOLOGY

Acute uncomplicated UTI manifesting as cystitis is a common syndrome that affects otherwise healthy women.^{1,3} About 10% of young, sexually active, premenopausal women experience a UTI each year, and 60% of all women have one or more such infections in their lifetimes. From 2% to 5% of women experience frequent recurrent infection for at least some period. After a first episode of cystitis, 21% of female college students in one study reported a second infection within 6 months.³⁷ Among postmenopausal women aged 55 to 75 years who were enrolled in a Seattle health group,

Table 38.3 Urinary Excretion of Antimicrobial Agents in Persons with Normal Renal Function

Antimicrobial Agent ^a	% Of Absorbed Drug Excreted Renally as Parent Metabolites (Active Metabolites) ^b
Penicillins	
Penicillin G	80
Amoxicillin	90
Amoxicillin/clavulanic acid	Clavulanate: 20-60
Ampicillin	90 (10)
Cloxacillin	35-50
Piperacillin	50-80
Piperacillin/tazobactam	Tazobactam: 60-80
Pivmecillinam	45
Cephalosporins	
Cephalexin	>80 (18)
Cefazolin	>80
Cefuroxime	>80
Cefotaxime	50-60 (30)
Ceftriaxone	50
Cefepime	85
Cefixime	15-20
Cefpodoxime	20-35
Cefprozil	60
Ceftazidime	80-90
Ceftaroline	50
Ceftolozane/tazobactam	95/80
Ceftazidime/avibactam	80-90/97
Macrolides, Lincosamides	
Erythromycin	5-15
Clindamycin	≤6 (some active metabolites)
Clarithromycin	20-30 (10-15)
Azithromycin	6
Aminoglycosides	
Gentamicin	99
Tobramycin	99
Amikacin	99
Carbapenems	
Imipenem/cilastatin	70-76
Meropenem	70-80
Ertapenem	40
Doripenem	70 (15)
Fluoroquinolones	
Norfloxacin	25-40 (10-20)
Ciprofloxacin	40 (10-20)
Levofloxacin	70-80
Moxifloxacin	20
Other Antibacterials	
Vancomycin	>90
Teicoplanin	>90
Dalbavancin	42
Daptomycin	54
Linezolid	35
Tigecycline	32

Continued on following page

Table 38.3 Urinary Excretion of Antimicrobial Agents in Persons with Normal Renal Function (Cont'd)

Antimicrobial Agent ^a	% Of Absorbed Drug Excreted Renally as Parent Metabolites (Active Metabolites) ^b
Colistin	64-70
Trimethoprim	66-95
Sulfamethoxazole	20-40
Nitrofurantoin monohydrate/macrocrystals	40-60
Fosfomycin tromethamine	30-60
Doxycycline	20-30
Aztreonam	66
Metronidazole	15 (30-60)
Rifampin	<10/50
Antifungals	
Amphotericin B deoxycholate	<10
Amphotericin B lipid formulations	<1
5-Flucytosine	90
Ketoconazole	<10
Fluconazole	80
Itraconazole	<1
Voriconazole	<1
Posaconazole	<1
Caspofungin	<1
Micafungin	<1
Anidulafungin	<1

^aNot all antimicrobial agents have an indication for urinary tract infection.

^bExcept where noted, values are the proportion of dose renally excreted unchanged.

the incidence was seven infections per 100 patient-years; in 24 months, 7% of women had one infection, 1.6% had two infections, and 1% had three or more infections.³⁸ Acute cystitis is associated with considerable short-term morbidity.³⁹ Female college students reported that symptoms persisted for an average of 6.1 days,² and, for ambulatory women, a mean duration of symptoms was 4.9 days, with 63% of patients reporting their usual activities were compromised by the infection.⁴⁰ However, although a large number of women are affected and many have frequent recurrence of infection, there is no observed long-term morbidity. Acute, uncomplicated UTI is uncommon in healthy young men, with an estimated incidence of <0.1% per year.

PATHOGENESIS AND MICROBIOLOGY

Acute UTI is primarily a disease of extraintestinal (or “uropathogenic”) *E. coli*. These organisms are isolated in 80% to 85% of episodes of acute cystitis.^{1,2,38} Infection occurs via the ascending route after bacterial strains originating in the gut colonize the vagina or periurethral area.¹¹ Although urethral colonization with a potential uropathogen appears to be a prerequisite for infection,

cystitis does not subsequently develop in most women with periurethral colonization.¹¹ Strains of *E. coli* that colonize the periurethral area and subsequently cause UTI belong to a restricted number of phylogenetic *E. coli* groups and more frequently express diverse virulence factors.^{41,42} A necessary characteristic for bladder infection is production of FimH, an adhesin attaching to receptors on uroepithelial cells.⁴³ This surface protein, however, is common on *E. coli* strains, regardless of whether they cause infection. Other potential urovirulence characteristics include other adhesins, iron-sequestration systems, toxin secretion, and motility.⁴² The putative virulence factors produced by strains isolated from symptomatic infection overlap with those isolated from asymptomatic bacteriuria, and no single characteristic is uniquely correlated with symptomatic infection.⁴⁴ Uropathogenic *E. coli* strains may be acquired during travel, from ingestion of food or water, from other household members including pets, and from sexual partners.^{45,46} Clonal outbreaks may occur with transmission of a single strain within a community or larger geographic area.^{47,48}

Other Enterobacteriaceae, most commonly *Klebsiella pneumoniae* and *Proteus* spp., are isolated in less than 5% of premenopausal women but in 10% to 15% of postmenopausal women.^{38,49}

Historically, *Staphylococcus saprophyticus* was hailed in some published series as an important uropathogen causing 5% to 10% of episodes, particularly in younger, sexually active women.⁵⁰ Yet many geographic regions report little if any *S. saprophyticus* in women with active infection.^{34,51} One report describes urogenital and rectal colonization in 7% and 40% of healthy women, respectively,⁵² leading to the question of whether this organism may primarily be a urinary contaminant with robust growth in traditional culture. Nonetheless, given its possession of virulence factors including adhesins, transport systems to support growth in urine, and urease production, a potential for uropathogenicity cannot be ruled out.⁴⁵

Other gram-positive organisms such as *Enterococcus* species and group B streptococci are uncommon pathogens in patients without foreign body material, though they are often present in midstream-voided specimens.³⁴ *Salmonella* species and bacteria associated with sexually transmitted infections, such as *Ureaplasma urealyticum*, *Gardnerella vaginalis*, and *Mycoplasma hominis*, are occasionally isolated.⁵³

HOST FACTORS

Acute, uncomplicated UTI is a consequence of the interaction of a virulent organism with host genetic susceptibility and behavioral variables.^{15,19} The notion of genetic propensity is supported by two consistent observations: (1) an increased frequency of UTI in first-degree female relatives of women with recurrent infection^{1,3,54} and (2) the fact that infection at a younger age is a major risk factor for recurrent cystitis in women of any age.^{1,38,54} One well-characterized genetic association is being a nonsecretor of the ABO blood-group antigens.^{54,55} Women with recurrent UTI are at least three times more likely to be nonsecretors than are those without recurrent infection. Nonsecretors express cell-surface glycosphingolipids on the vaginal epithelium, and presumably urethral mucosa, that differ from those

expressed by secretors and bind uropathogenic *E. coli* more avidly.¹⁹ Other potential genetic determinants include polymorphisms of the IL-8 receptor CXCR1, TLRs, and the tumor necrosis factor promoter.¹⁹

The most important behavioral association of UTI in premenopausal women is sexual intercourse.^{1-3,56} In young, sexually active women, 75% to 90% of episodes are attributable to intercourse, and there is a correlation between frequency of intercourse and frequency of infection.⁵⁵ Intercourse appears to promote infection by facilitating ascension of organisms from the periurethral area into the bladder, but disruption of the female uromicrobiota after interaction with that of her partner may also play a role.¹⁴ Spermicide use is another independent behavioral risk factor for acute cystitis in premenopausal women.⁵⁵ Spermicides are bactericidal for the hydrogen peroxide-producing lactobacilli of the normal vaginal microbiota, which maintain the acidic pH. When these bacteria are not present, vaginal pH increases, facilitating colonization with potential uropathogens. Case-control studies have consistently demonstrated that behavioral variables popularly identified as risks for cystitis—such as type of underwear, bathing rather than showering, postcoital voiding, frequency of voiding, perineal hygiene practices, vaginal douching, and tampon use—are not associated with an increased risk of infection.^{1,2}

A history of prior UTI at a younger age is the strongest association of recurrent acute cystitis in postmenopausal women.⁵⁷⁻⁵⁹ Sexual intercourse is not an important contributor in this population.^{59,60} Estrogen deficiency has been proposed to promote recurrent UTI in these women through alterations in vaginal flora, including replacement of lactobacilli by potential uropathogens. However, prospective cohort studies and case-control studies uniformly demonstrate no association of oral or topical estrogen use with recurrent UTI, regardless of restoration of vaginal lactobacilli and acid pH.⁶¹ Acute uncomplicated UTI in men is uncommon, but reported risk factors have included intercourse with a female partner with recurrent UTI, not being circumcised, and anal intercourse.⁶²

DIAGNOSIS

The classic clinical manifestation of symptomatic lower UTI is the acute onset of one or more irritative bladder symptoms, such as urgency, frequency, dysuria, stranguria, and hesitancy.^{1,3} Gross hematuria may also occur, likely reflecting more severe mucosal inflammation. The most important differential diagnoses to exclude are sexually transmitted infections⁶³; vulvovaginal candidiasis and noninfectious syndromes such as interstitial cystitis should also be considered. The combination of new-onset frequency, dysuria, and urgency, together with the absence of vaginal discharge and pain, has a positive predictive value for acute cystitis of 90%.⁶⁴ Women with recurrent UTI also have >90% accuracy in self-diagnosis on the basis of symptoms.²

A urine culture is not recommended routinely for women with a clinical presentation consistent with acute uncomplicated cystitis given a high pretest probability of infection, predictable microbiology, and prompt clinical response with empirical antimicrobial therapy.^{3,53,65} Urine culture results are negative in up to 20% of women with

a characteristic clinical presentation, and these women have a clinical response to antimicrobial therapy similar to that of women with positive cultures.⁶⁶ These women with low organism counts isolated in urine culture may have urethritis, rather than cystitis. However, both urinary frequency and increased fluid intake are characteristic of patients with acute cystitis; limited dwell time and/or dilution of urine in the bladder seem the likely explanation for lower bacterial counts. Finally, fastidious organisms may not be identified by routine laboratory procedures used in processing urine specimens.

A urine specimen for culture should be obtained, however, in certain instances. When the clinical presentation is not characteristic, a urine culture may help confirm or exclude the diagnosis of UTI. Any quantitative count of an Enterobacteriaceae is considered positive. *Enterococcus* spp. or group B streptococcus in any quantitative count should be interpreted as contamination.³⁴ Failure to respond to appropriate empirical therapy or early (<1 month) recurrence after therapy is suggestive of infection with a resistant organism. In these situations, a urine culture should be obtained to confirm resistance and facilitate selection of an effective regimen.

The presence of pyuria, identified by routine urinalysis or leukocyte esterase dipstick testing, is a consistent accompaniment of acute cystitis.^{1,65} The absence of pyuria is suggestive of an alternative diagnosis but does not rule out UTI in women with a consistent clinical presentation, particularly given increasing rates of self-hydration among women before seeking medical attention for lower UTI.^{1,67-69} Thus routine screening for pyuria is not strictly necessary for diagnosis or management of acute cystitis.^{1,65} A urine nitrite dipstick test screens for the presence of bacteria, rather than leukocytes, and results are usually positive in women with infection.⁶⁹ False-negative nitrite test results may occur when there is infection with bacteria that do not reduce nitrate, such as *Enterococcus* species, or when urine has not been retained in the bladder a sufficient time to allow bacteria to convert nitrate to nitrite. Nitrite tests uncommonly have false-positive results, but these may occur when blood, urobilinogen, or some dyes are present in the urine.

TREATMENT

For many women, acute uncomplicated cystitis may spontaneously resolve both clinically and microbiologically within a few days or weeks. In a clinical trial in which participants were randomly assigned to antibiotic therapy or placebo, 28% of 277 women who received placebo were asymptomatic by 1 week, and 45% had negative culture results by 6 weeks.⁷⁰ In another study, 54% of women who received placebo were asymptomatic by 3 days and 52% at 7 days.⁷¹ Antimicrobial treatment, however, is associated with significantly shorter symptom durations. The rates of clinical cure were 77% with nitrofurantoin versus 54% with placebo at 3 days and 88% and 52%, respectively, at 7 days.⁷¹ After initiation of antimicrobial therapy, 54% of women reported symptom improvement by 6 hours, 87% by 24 hours, and 91% by 48 hours. In another case series, 72% of women reported complete symptom resolution by the fourth day of effective treatment.⁴⁰ In a trial of antiinflammatory therapy with ibuprofen alone compared

with empiric fosfomycin therapy for women with mild-to-moderate symptoms, women receiving fosfomycin had more rapid symptom resolution and fewer cases of pyelonephritis, though these were few in both arms.⁷²

Many antimicrobial agents are effective for treatment of acute cystitis (Table 38.4). The anticipated cure rate for recommended first-line empirical regimens is 70% to 95%.^{3,66,73} While TMP/sulfamethoxazole (TMP/SMX) was a mainstay of empirical treatment of acute cystitis for decades and remains highly effective against susceptible organisms,^{73,74} increasing rates of TMP/SMX resistance in community *E. coli* isolates compromised its use as first-line empirical therapy.⁷⁵ As a result, older antibacterial agents have been repositioned as first-line therapy: nitrofurantoin, pivmecillinam, and oral fosfomycin have indications restricted to acute cystitis. These medicines do not induce cross-resistance with other classes of antimicrobial agents and, to date, limited resistance has been observed in community uropathogens.^{3,75} In particular, despite high nitrofurantoin consumption worldwide in the past decade, *E. coli* strains remain overwhelmingly susceptible, likely due to nitrofurantoin's multiple mechanisms of action.⁷⁶

Thus they are ecologically attractive for treatment of acute cystitis.^{39,73,77} Pivmecillinam, however, is not available in many countries, and in countries where fosfomycin is also used parenterally, resistance rates to this antibiotic are increasing.⁷⁸ The fluoroquinolones with urinary excretion—norfloxacin, ciprofloxacin, and levofloxacin—are no longer recommended as first-line therapy because widespread use promotes the emergence of resistance, particularly among Enterobacteriaceae.⁷⁷ Oral β -lactam antimicrobial agents including amoxicillin, amoxicillin/clavulanic acid, and cephalosporins are reported to be 10% to 15% less effective than first-line agents.^{73,79} The β -lactam agents are useful, however, for treatment in pregnant women given their favorable safety profile.⁵³

To limit adverse effects, cost, and emergence of bacterial resistance, the shortest effective duration of antimicrobial therapy should be prescribed. For TMP/SMX and the fluoroquinolones, 3 days is optimum.^{3,73} The minimum duration of nitrofurantoin therapy is 5 days.^{76,80} For β -lactam antimicrobial agents, 7 days is recommended; shorter courses are less effective. Fosfomycin is approved only as single-dose therapy, though a single dose is clinically and microbiologically inferior to 5 days of nitrofurantoin.⁶⁶ For other agents, a single dose is generally 5% to 10% less effective than recommended longer regimens.^{53,73}

Antimicrobial susceptibility of uropathogenic *E. coli* strains acquired in the community evolves continually in response to antimicrobial pressure.^{75,81-83} Resistance in community isolates has compromised the efficacy of ampicillin, cephalosporins, TMP/SMX, and fluoroquinolones.^{3,75} Recent prior antimicrobial therapy is a strong risk factor for isolation of a resistant organism.⁸³ If the local prevalence of resistance to an antimicrobial agent in community *E. coli* strains exceeds 20%, that agent should not be used for first-line empirical therapy.⁷³ The current increase in extended-spectrum β -lactamase (ESBL)-producing *E. coli* in community-acquired infections, attributed to global expansion of the *E. coli* clone ST131,⁴⁸ is of particular

Table 38.4 Preferred Antimicrobial Regimens for Treatment or Prevention of Acute, Uncomplicated Urinary Infection (Cystitis) in Women with Normal Renal Function

First-Line Therapy	Other Therapy
Acute Cystitis	
Nitrofurantoin monohydrate/macrocrystals 100 mg tid × 5 days ^{a,b}	Amoxicillin 500 mg tid × 7 days ^a
(If microcrystalline formulation, 50-100 mg qid) ^c	Amoxicillin/clavulanic acid, 500 mg tid or 875 mg bid × 7 days ^a
Pivmecillinam, 400 mg bid × 5 days or 200 mg bid × 7 days ^a	Cephalexin, 250-500 mg qid × 7 days ^a
Fosfomycin trometamol, 3-g single dose ^a	Cefpodoxime proxetil, 100 mg bid × 3 days
TMP/SMX, 160/800 mg bid × 3 days ^d	Cefuroxime axetil, 500 mg bid × 7 days ^a
	Cefixime, 400 mg qd × 7 days ^a
	Norfloxacin, 400 mg bid × 3 days
	Ciprofloxacin 500 mg bid
	Levofloxacin, 500 mg qd × 3 days
	Trimethoprim, 100 mg bid × 3 days ^d
	Doxycycline, 100 mg bid × 7 days
Prophylaxis	
Long-Term Low-Dose Regimens (at Bedtime)	
Nitrofurantoin monohydrate/macrocrystals, 50 mg or 100 mg qd	Cephalexin, 500 mg qd ^a
	TMP/SMX, 40/200 mg od or every other day ^d
	Trimethoprim, 100 mg qd ^d
Postcoital (Single-Dose) Regimen	
Nitrofurantoin, 50 or 100 mg ^a	Cephalexin, 250 mg ^a
	TMP/SMX, 40/200 mg or 80/400 mg ^d
	Trimethoprim, 100 mg ^d

^aRecommended for pregnant women.

^bNitrofurantoin (in monohydrate/macrocrystalline form) may also be given at a dose of 100 mg bid, but tid dosing may be more effective given nitrofurantoin's time-dependent activity against *Escherichia coli* strains.³⁶

^cMicrocrystalline nitrofurantoin is still available but is not recommended first-line therapy because of its higher rate of gastrointestinal side effects.³⁵¹

^dTMP/SMX use should be avoided in regions where uropathogen resistance rates are $\geq 20\%$.

bid, Twice a day; qd, once a day; qid, four times a day; tid, twice

concern because these strains are usually also resistant to TMP/SMX and fluoroquinolones.^{83,84} Optimal treatment of infection with ESBL-producing *E. coli* is not yet well defined. Nitrofurantoin, pivmecillinam, and fosfomycin currently remain effective for most of these strains.^{83,85}

OTHER INVESTIGATIONS

Young women with a characteristic clinical presentation and prompt response following appropriate empirical therapy do not require further diagnostic imaging or urologic investigation.⁸⁶ Less than 5% of these women have urologic abnormalities, and the few women in whom abnormalities are identified usually do not require further intervention. However, investigation may be appropriate to rule out alternative pathologic processes when the diagnosis is uncertain or clinical presentation is atypical.

PREVENTION AND MANAGEMENT OF RECURRENT INFECTION

Management strategies for recurrent lower UTI range from nonpharmacologic approaches to daily antibiotic prophylaxis; given perpetually increasing antimicrobial resistance rates, daily antibiotic prophylaxis should be considered last resort and initiated only when underlying risk factors cannot be corrected or mitigated.

Increased daily hydration can significantly reduce incidence of recurrent UTI. A randomized clinical trial comparing increased water intake (1.5 L added to baseline daily water intake) compared with usual behavior in otherwise healthy women with recurrent UTI who at baseline took in ≤ 1.5 L of water resulted in nearly half as many UTI episodes over the 12-month study period.⁸⁷

Other proposed nonantimicrobial approaches for prevention include daily intake of cranberry products, oral or vaginal probiotics to reestablish normal vaginal flora, estrogen replacement for postmenopausal women, and oral or bladder-instilled hyaluronic acid with chondroitin sulfate to reinstate the glucosaminoglycan layer.^{88,89} Initial studies reported that daily intake of cranberry juice or tablets decreased episodes of recurrent cystitis by 30% in comparison with placebo.⁹⁰ The proposed mechanism for this effect is inhibition of the P fimbria-mediated adherence of *E. coli* to uroepithelial cells by the proanthocyanidins present in cranberry products and excreted in the urine. However, more recent clinical trials report no benefit of daily cranberry juice in comparison with placebo for prevention of active infection⁹¹ or bacteriuria,⁹² and cranberry capsules were less effective than TMP/SMX prophylaxis.⁹³ Though there seems to be a protective effect of host *Lactobacillus* spp. in some epidemiologic studies,¹⁴ trials of oral or vaginal *Lactobacillus* probiotics do not support efficacy of these products in preventing UTI.⁹⁴⁻⁹⁶ The role of estrogen replacement to prevent urinary tract infection in postmenopausal women is controversial.^{2,97} In prospective clinical trials, researchers have uniformly reported no benefit of systemic estrogen over placebo, despite restoration of an acidic vaginal pH and increased vaginal colonization with lactobacilli in women who received estrogen.⁶¹ Two small clinical trials comparing vaginal estrogen with placebo in postmenopausal women with frequent, recurrent infection demonstrated a decreased frequency of symptomatic episodes in women treated with topical estrogen therapy, but in a comparative trial, an estrogen-containing pessary was substantially less effective than nitrofurantoin prophylaxis.^{61,98} Currently, topical vaginal estrogen should likely only be considered for use to prevent infection in selected women with a high frequency of recurrent infection. Finally, trials have not demonstrated

efficacy of either oral or bladder-instilled hyaluronic acid and chondroitin sulfate.^{99,100}

Several other nonantimicrobial approaches for prevention of recurrent UTI remain under investigation.^{4,101,102} These include strategies to block bacterial FimH adhesion with D-mannose,¹⁰² using vitamin D to enhance antimicrobial peptide production,¹⁰³ vaccination using FimH or iron receptors,¹⁰² and immune stimulation with heat-killed whole bacteria.¹⁰² Trials with one of the whole bacterial immunostimulants (OM-89S), however, have shown no benefit.¹⁰⁴ The clinical efficacy of any of these proposed interventions remains uncertain.

In women whose recurrent UTI is linked to sexual activity, postcoital antibiotic prophylaxis can significantly reduce UTI episodes.^{105,106} To reduce emergence of antimicrobial resistance, however, only nitrofurantoin (50 mg or 100 mg as a single dose) should be prescribed given its minimal collateral effects on the intestinal microbiota.¹⁰⁷

Likewise, in patients fulfilling criteria for daily antibiotic prophylaxis, which are an inability to correct or mitigate underlying risk factors and the occurrence of two UTI episodes in the previous 6 months or three episodes in the previous year, nitrofurantoin (50 or 100 mg taken at bedtime) is the first-line agent, as regular intake of other antibiotics such as fluoroquinolones or TMP/SMX will result in rapid emergence of resistant Enterobacteriaceae. Daily prophylaxis should be prescribed initially for 3 or 6 months at most, with regular reevaluation of the need for prophylaxis. Patients and their primary care physicians should be advised that long-term daily nitrofurantoin administration has been associated with rare occurrences of pulmonary or hepatic hypersensitivity, since tissue fibrosis may ensue if nitrofurantoin is not discontinued.¹⁰⁸ The incidence of hypersensitivity is exceedingly rare, however, and this event has almost never been observed in patients taking short courses (≤ 14 days) of nitrofurantoin since the medicine's introduction to the market in 1953.^{76,108}

Self-treatment ("pill in the pocket") is an effective strategy for managing recurrent infections, especially given women's accuracy in self-diagnosing UTI.² While a 3-day course of TMP/SMX or ciprofloxacin has been shown to be effective for empirical self-treatment in clinical trials, self-treatment with these agents is no longer recommended due to resistance development. Instead, a 5-day course of nitrofurantoin 100 mg taken three times daily can be prescribed.⁶⁶ In the event of nitrofurantoin intolerance or resistance, single-dose oral fosfomycin may be tried.

COMPLICATED URINARY TRACT INFECTION : ACUTE NONOBSTRUCTIVE PYELONEPHRITIS

EPIDEMIOLOGY

Pyelonephritis is much less common than cystitis. The ratio of pyelonephritis to cystitis episodes is reported to be between 1:18 and 1:29 in women with recurrent infection.² The highest incidence is among women aged 20 to 30 years. Pyelonephritis is associated with substantial morbidity; hospitalization is required for up to 20% of affected nonpregnant women.¹⁰⁹ Severe manifestations such as sepsis syndrome are, however, uncommon. Acute pyelonephritis complicates 1% to 2% of pregnancies. When this complication occurs at the end of the second trimester

or early in the third trimester, preterm labor and delivery may occur and lead to poor fetal outcomes, as with any febrile illness in later pregnancy.¹¹⁰

Acute nonobstructive pyelonephritis is rarely a direct cause of renal failure. Renal scarring is a complication of pyelonephritis in some women with more severe clinical presentations. In an Israeli study, 30% of a cohort of 203 women admitted for acute pyelonephritis were reassessed 10 to 20 years after admission; renal scars were detected in 46% of these women on technetium (Tc) 99m-labeled dimercaptosuccinic acid scanning. However, the renal scars were not associated with hypertension or renal impairment.¹¹¹ The histologic finding of “chronic pyelonephritis” in patients with renal failure was formerly attributed to infection. However, this condition is now recognized as an end stage of many chronic inflammatory conditions of the kidney, and it is attributable to infection in only a few patients in whom there is a clear history of recurrent renal infection.

PATHOGENESIS

E. coli is isolated in 85% to 90% of women who present with acute pyelonephritis.^{2,112} Infecting strains are characterized by production of the P fimbria adhesin Gal(α 1–4) Gal β disaccharide galabiose. This surface protein appears to have a direct role in the pathogenesis of pyelonephritis through induction of mucosal inflammation.^{42,113} A familial susceptibility to pyelonephritis has been reported and attributed to polymorphisms with decreased expression of the IL-8 receptor.¹¹⁴ Other genetic and behavioral risk factors for pyelonephritis in healthy women are similar to those described for acute cystitis.¹¹² For premenopausal women, these associations are frequency of sexual intercourse, history of UTI, history of UTI in the patient’s mother, a new sexual partner, and recent spermicide use. The strongest association is with recent sexual intercourse. Women who experience pyelonephritis as children remain at increased risk of pyelonephritis as adults but experience a lower frequency of episodes.¹¹⁵ Diabetes is also an independent risk factor for pyelonephritis. Young diabetic women are 15 times more likely to be hospitalized for pyelonephritis than age-matched nondiabetic women. Behavioral risk factors associated with pyelonephritis in postmenopausal women have not yet been identified.

DIAGNOSIS

The classic clinical manifestation of renal infection is costovertebral angle pain or tenderness, often accompanied by fever, with or without a prodrome of lower urinary tract symptoms. There is a wide spectrum of severity, however, from mild irritative symptoms with minimal costovertebral angle tenderness to severe symptoms that may include high fever, nausea and vomiting, and severe pain. Acute cholecystitis, renal colic, and pelvic inflammatory disease (PID) are occasionally confused with pyelonephritis. When patients present with severe symptoms, underlying complicating factors such as obstruction and abscess must be excluded through urgent imaging. A urine specimen for culture should be obtained before initiation of antimicrobial therapy in every case of suspected pyelonephritis to confirm diagnosis and guide therapy.

Bacteremia is identified in 10% to 25% of women presenting with acute pyelonephritis if blood culture specimens are collected routinely. However, the clinical utility of routine blood cultures is limited because bacteremia does not alter therapy, nor is it predictive of outcome.^{116,117} They should thus be obtained selectively, typically only if the diagnosis is uncertain or the presentation is severe.

Additional investigations recommended for most patients presenting with acute pyelonephritis are measurements of peripheral leukocyte count, C-reactive protein (CRP), and serum creatinine level. The CRP is typically elevated within the first days of symptoms and is useful as a parameter to monitor the response to therapy; its peak value correlates with severity of presentation.¹¹⁸ Elevated CRP at discharge has been associated with prolonged hospitalization and postdischarge recurrence.¹¹⁸

DIAGNOSTIC IMAGING

Routine diagnostic imaging is not indicated when the clinical presentation is characteristic, symptoms are mild or moderate, and response to antimicrobial therapy is prompt.^{53,119} Patients whose clinical presentations are severe, in whom treatment fails, or who experience early posttreatment recurrence should undergo prompt imaging to rule out obstruction or abscesses and to determine whether intervention is necessary. Ultrasonography is often the initial imaging modality because it is safe and widely accessible.¹²⁰ In women with pyelonephritis and no preexisting structural abnormalities, it usually yields normal results, but enlargement and edema in one or both kidneys are observed in 20% of patients.¹²¹ Ultrasonography is less sensitive or specific for pyelonephritis than is either computed tomography (CT) or magnetic resonance imaging (MRI).

The optimal diagnostic imaging is contrast-enhanced CT, though this may be associated with a risk of developing contrast-induced nephropathy.¹²¹ In addition to renal enlargement and edema, dilation of the collecting system in the absence of obstruction, wedge-shaped areas of decreased attenuation, and rounded low-attenuation masses with delayed enhancement may be observed. Obstruction of renal tubules by inflammatory debris or impaired function with tubular ischemia may result in a “striated nephrogram.”¹²⁰ Abnormalities within the renal cortex and medulla, inflammatory changes in Gerota fascia or the renal sinuses, and thickening of the urothelium are sometimes observed. “Acute focal pyelonephritis” (also called “acute lobar nephronia”) is infection confined to a single lobe and may be more common in women who have diabetes or are immunocompromised. The response to treatment is similar, however, for patients with or without this finding.¹²¹

TREATMENT

Most women with pyelonephritis receive treatment as outpatients.^{53,73} Indications for hospitalization include hemodynamic instability, the need to exclude complicating factors such as obstruction and abscess, and the necessity of monitoring or treatment of associated medical illnesses. Traditionally, uncertain gastrointestinal absorption and pregnancy were also indications for hospitalization, but increased availability of outpatient parenteral therapy with close outpatient clinical follow-up may obviate the need for it.

Appropriate supportive management for hypotension, nausea and vomiting, and pain should be initiated promptly. When oral tolerance is uncertain because of nausea and vomiting, a strategy frequently used in emergency department management is to provide a single parenteral dose of ceftriaxone, 1 g, or of gentamicin, 120 mg, followed by oral therapy once gastrointestinal symptoms are controlled.

Many parenteral antimicrobial regimens are effective for pyelonephritis (Table 38.5). Options for empiric parenteral therapy include third-generation cephalosporins such as cefotaxime and ceftriaxone, aminoglycosides, and fluoroquinolones such as ciprofloxacin.^{2,73} Aminoglycosides have unique efficacy for the treatment of renal infection as they are bound in high concentrations in the renal cortex.² Ceftriaxone therapy is the preferred empirical regimen for pregnant women. Although it is suggested that gentamicin be avoided in pregnancy because of potential fetal ototoxicity, excess otologic impairment has not been reported in large cohorts of newborn infants stratified by gentamicin exposure in utero.¹²² Thus when cephalosporins cannot be used because of antimicrobial resistance or patient intolerance, gentamicin remains an alternate antimicrobial for treatment of pregnant women. For ESBL-producing *E. coli* resistant to all first-line therapies, a carbapenem (imipenem/cilastatin, meropenem) is recommended.

Oral therapy selected according to culture results can then be prescribed to complete the antimicrobial course. In most young, nonpregnant women, acute pyelonephritis is effectively managed with outpatient oral therapy (Table 38.5).^{53,73} The recommended empirical antimicrobial choice is ciprofloxacin.⁷³ Oral TMP/SMX is effective, but because of the high prevalence of TMP/SMX-resistant *E. coli* in some areas, this agent is recommended only when the infecting organism is known to be susceptible. Other oral regimens are also effective and may be appropriate, depending on organism susceptibility and the patient's tolerance. Several trials have shown that 5- to 7-day antibiotic durations are noninferior to 14-day durations, even in patients with concomitant bacteremia.^{123,124} Ciprofloxacin 500 mg twice daily given for 5 or 7 days^{125,126} or a course of a single dose of ceftriaxone 1 g and cefixime 400 mg daily for 6 days¹²⁷ are effective shorter regimens.¹²⁸

A satisfactory clinical response is usually observed by 48 to 72 hours after initiation of antimicrobial therapy. Risk factors predictive of a poor outcome are hospitalization, isolation of a resistant organism, concurrent diabetes mellitus, and history of renal stones.¹²⁹ Prophylactic antimicrobial strategies similar to those for recurrent cystitis are effective for prevention of recurrent uncomplicated pyelonephritis.

COMPLICATED URINARY TRACT INFECTION: OBSTRUCTIVE PYELONEPHRITIS AND CATHETER-RELATED INFECTION

EPIDEMIOLOGY

Patients with indwelling devices or persistent obstruction may experience frequent recurrent infections. In men with spinal cord injury in whom voiding is managed with an indwelling catheter, UTI incidence is 2.72 infections/1000 days; when voiding is managed with intermittent catheterization, the infection incidence is 0.41/1000 days.¹³⁰

Table 38.5 Antimicrobial Regimens for Treatment of Acute, Uncomplicated Pyelonephritis in Women with Normal Renal Function and Susceptible Organisms

First-Line Therapy	Other Therapy
Oral	
Ciprofloxacin, 500 mg bid or 1000 mg extended-release preparation, qd × 5–7 days	TMP/SMX, 160/800 mg bid × 7 days
Levofloxacin, 750 qd × 5–7 days	Amoxicillin, 500 mg PO tid × 7 days ^a
	Amoxicillin/clavulanic acid, 500 mg tid or 875 mg PO bid × 7 days ^a
	Cephalexin, 500 mg qid × 7 days ^a
	Cefuroxime axetil, 500 mg bid × 7 days ^a
	Cefixime, 400 mg qd × 7 days ^a
Parenteral^b	
Ciprofloxacin, 400 mg q12h	Meropenem, 1 g q8h
Levofloxacin, 750 mg qd	Piperacillin/tazobactam, 3.375 g q6h
Gentamicin or tobramycin, 3–5 mg/kg qd, ± ampicillin, 1 g q4–6h	Doripenem, 500 mg q8h
Ceftazidime, 2 g q8h	Ertapenem, 1 g qd
Ceftriaxone, 1–2 g qd ^a	Ceftolozane/tazobactam, 1 g q8h
Cefotaxime, 1 g q8h ^a	Ceftazidime/avibactam, 2 g q8h

^aRecommended for pregnant women.

^bChange to oral therapy to complete course once condition is clinically stable.

bid, Twice a day; *PO*, by mouth; *qd*, once a day; *qid*, four times a day; *tid*, twice a day; *TMP/SMX*, trimethoprim/sulfamethoxazole.

In residents of long-term care facilities with long-term indwelling urethral catheters, the incidence of symptomatic infection is 3.2/1000 catheter days.¹³¹

These patients are also at risk for local suppurative complications, such as renal or perinephric abscess, or metastatic infection after bacteremia, such as septic arthritis, osteomyelitis, or endocarditis. Serious complications are more common in patients who are diabetic, are immunocompromised, or have long-term urologic devices or obstruction. Renal functional impairment in patients with complicated UTI is usually attributable to the underlying abnormality or to organ failure complicating septic shock, rather than being a direct consequence of infection. For instance, introduction of voiding strategies to maintain low bladder pressure and to prevent reflux have almost eliminated the complication of chronic renal failure in persons with spinal cord injury, despite a continued high incidence of UTI in these patients.¹³²

PATHOGENESIS AND MICROBIOLOGY

Host impairment rather than organism virulence is the major determinant of infection. *E. coli* remains the organism

most frequently isolated in complicated urinary tract infection.^{8,9} The *E. coli* strains are characterized by a low frequency of expression of virulence factors in comparison with strains isolated in uncomplicated infection.⁴⁴ Many other bacteria and yeast species are also isolated.^{9,133,134} Enterobacteriaceae such as *Klebsiella* species, *Enterobacter* species, *Serratia* species, *Citrobacter* species, *Proteus mirabilis*, *Morganella morganii*, and *Providencia stuartii* are common. Other gram-negative organisms that may be isolated include *Pseudomonas aeruginosa*, *Stenotrophomonas maltophilia*, and *Acinetobacter* species. Gram-positive organisms are also frequently isolated including group B streptococci, *Enterococcus* species, and coagulase-negative staphylococci. *S. aureus* is less commonly isolated. *Candida* species may be isolated, usually from patients who have diabetes or indwelling urologic devices or who are undergoing broad-spectrum antimicrobial therapy.¹³⁵ Organisms isolated from patients with complicated UTI often have increased antimicrobial resistance.¹³⁶ Risk factors for isolation of a resistant organism include a history of recent antimicrobial therapy or hospitalization, indwelling urethral catheter, invasive urologic procedures, and nursing home residence.

Biofilm formation is universal in patients with chronic indwelling urinary devices.¹³⁷ Following insertion, a conditioning layer composed of proteins and other host components immediately coats the device. This conditioning layer provides a surface for subsequent attachment of bacteria or yeast that originate from the periurethral flora or drainage bags or are introduced after disruption of the closed drainage system. Organisms grow along the device, elaborating an extracellular polysaccharide substance, and colonies of microorganisms persist within this relatively protected environment. Organisms ascend in the biofilm along the interior and exterior surfaces of the device and reach the bladder within days. The initial infection is usually with a single organism, but a polymicrobial flora is invariably present on long-term indwelling devices.^{31,137} Urease-producing organisms such as *P. mirabilis*, *K. pneumoniae*, *M. morganii*, and *P. stuartii* are isolated more frequently from individuals with long-term indwelling catheters and persist longer than other organisms, such as *Enterococcus* species. Complications attributed to biofilm formation with urease-producing organisms include development of renal or bladder stones and obstruction of the device.¹³⁸

Although the most common device is the urethral catheter, the process of biofilm formation is similar with other indwelling devices, such as ureteric stents and nephrostomy tubes.¹³⁹⁻¹⁴¹ From 34% to 42% of ureteral stents are found at removal to be colonized with bacteria or yeast species, often with multiple organisms. More than 50% of organisms identified by stent biofilm analysis are not isolated from culture of urine specimens collected at the same time.

Uncommon bacteria are occasionally a cause of infection. Some of these organisms may not be identified with standard laboratory procedures for processing urine specimens. *Corynebacterium urealyticum* is a urease-producing gram-positive rod associated with the unique clinical manifestations of encrusted cystitis or pyelonephritis.¹⁴² This infection is characterized by ulcerative inflammation and struvite encrustations on the bladder or renal pelvis wall. Pyelitis, if untreated, may lead to destruction of

the kidney. *U. urealyticum* is another urease-producing bacterium that may cause cystitis or pyelonephritis, often with urolithiasis. Case reports suggest a predisposition in immunocompromised individuals, particularly those with hypogammaglobulinemia; healthy persons may also become infected. *Aerococcus sanguinicola* is a rare cause of complicated urinary tract infection; the diagnosis is usually made by isolation of the organism in blood culture.¹⁴³ *Aerococcus urinae* was isolated in 0.3% to 0.8% of urine specimens in one clinical microbiology laboratory, usually from older persons with underlying abnormalities and bacteremia.¹⁴⁴ Anaerobic organisms are seldom identified in the absence of suppurative complications such as abscess.¹⁴⁵

HOST FACTORS

Genitourinary abnormalities facilitate infection through increased entry of organisms into the bladder (e.g., by intermittent catheterization and urologic procedures) and subsequent persistence due to incomplete voiding (e.g., as a result of obstruction, urolithiasis, diverticula, and reflux), or in biofilm on urologic devices.^{8,9} Asymptomatic bacteriuria is common in patients with persistent abnormalities and universal in patients with long-term indwelling catheters.¹⁴⁶ The determinants that lead to symptomatic infection in chronically bacteriuric individuals are not well characterized. However, obstruction and mucosal trauma with bleeding are well-recognized risk factors for bacteremia and sepsis in patients with preexisting bacteriuria.

CLINICAL PRESENTATIONS

Obstructive and catheter-related complicated UTIs manifest across a wide clinical spectrum of signs and symptoms, from mild, irritative symptoms of lower-tract infection to pyelonephritis and bacteremia including septic shock.^{8,9,147} Localizing signs and symptoms consistent with cystitis or pyelonephritis are usually present. Patients with indwelling urethral catheters or other indwelling devices usually present with fever alone, although costovertebral angle pain or tenderness, hematuria, or catheter obstruction, if present, identifies a genitourinary source. Patients with chronic neurologic impairment sometimes report symptoms that are not classic for UTI. For instance, spinal cord-injured patients experience increased bladder and leg spasms or autonomic dysreflexia,¹³² whereas patients with multiple sclerosis may present with fatigue or deterioration in neurologic function.⁹

Clinical Diagnosis in Elderly Populations with Cognitive Impairment

The clinical diagnosis of symptomatic infection in older populations with cognitive impairment is often problematic.¹⁴⁸⁻¹⁵⁰ These patients frequently have chronic genitourinary symptoms and impaired communication, which limits the assessment of signs and symptoms. Because bacteriuria is common in elderly individuals with functional impairment, nonlocalizing clinical deterioration is frequently attributed to UTI because the urine culture is positive.¹⁵¹ However, nonlocalizing clinical manifestations including fever are unlikely to have a urinary source in elderly persons without a long-term indwelling catheter.^{30,149} Changes in character of the urine, such as cloudiness and odor, are also frequently interpreted as

symptoms of UTI. Cloudiness may be attributed to pyuria, which usually accompanies bacteriuria, and an unpleasant odor is suggestive of production of polyamines by bacteria in the urine. However, alterations in characteristics of the urine are neither sensitive nor specific for the diagnosis of infection. They may be attributable to other causes, such as precipitation of crystals and dehydration. Thus changes in character of the urine should not be interpreted as symptoms of UTI.

LABORATORY DIAGNOSIS

A urine specimen for culture should be obtained before initiation of antimicrobial therapy for every patient; given the wide variety of potential infecting organisms and increased likelihood of resistant strains, microbiologic characterization is necessary to optimize antimicrobial management.

The presence of biofilm on devices such as indwelling catheters and stents may complicate interpretation of culture results.^{11,137-139} The catheter should be replaced, and the new catheter should be used to sample bladder urine and avoid contamination by organisms present in the biofilm of the old catheter.^{30,31} Renal function should be assessed in every patient with complicated UTI. A second urine culture after antimicrobial therapy is not recommended if the patient remains asymptomatic.

Urine Culture in Patients with Ileal Conduits or Other Diversions

When the intestine is interposed into the urinary tract through creation of an ileal conduit, continent cutaneous diversion, or neobladder, urine collected through the conduit or reservoir is often bacteriuric regardless of symptoms.¹⁵² The organisms isolated represent mixed gram-positive flora including streptococcal species and *Staphylococcus epidermidis*, but uropathogenic strains such as *E. coli*, *P. mirabilis*, *P. aeruginosa*, and *Enterococcus faecalis* may also be present. Similar urine culture findings are reported in 30% to 60% of patients with orthotopic bladder substitution or augmentation cystoplasty¹⁵³; individuals with these reservoirs who practice clean intermittent catheterization are more likely to have positive culture findings.¹⁵² Thus when symptomatic UTI is suspected, results of a urine culture in such a patient must be interpreted in the context of this usual bacteriuria.

Patients with genitourinary abnormalities frequently have pyuria, whether or not they have bacteriuria or symptomatic infection, and so pyuria by itself is not diagnostic for urinary tract infection.⁹ The absence of pyuria, however, has a high negative predictive value for ruling out urinary tract infection in some populations, such as elderly patients.³⁰ The severity of clinical manifestations determines whether additional investigation, such as blood culture or a peripheral leukocyte count, is indicated.

IMAGING

The approach to diagnostic imaging is determined by the clinical presentation and the patient's previous history. Urgent imaging or urologic investigation is indicated for patients who have severe systemic symptoms or whose symptoms do not respond to appropriate antimicrobial therapy. The goal of early imaging is to identify obstruction

or abscesses, for which immediate drainage may be necessary for source control.¹³⁰ When the underlying abnormality is already well characterized (e.g., indwelling catheter and neurogenic bladder), further investigations may still be appropriate if infections have changed in frequency, as risks for urolithiasis and suppurative complications are elevated.

While a plain radiograph of the abdomen may identify emphysematous infections and some stones, CT is the imaging modality of choice. It identifies calculi, gas, hemorrhage, calcification, obstruction, renal enlargement, and inflammatory masses. Contrast media-enhanced scans are recommended (though are associated with risk of contrast-induced nephropathy), with helical and multislice CT to study different phases of contrast excretion.¹¹⁹⁻¹²¹ Ultrasound examination is less sensitive and specific than other imaging modalities such as spiral CT and MRI but may be more accessible.¹²¹

ANTIMICROBIAL TREATMENT

The antimicrobial regimen selected is individualized on the basis of site of infection, severity of manifestations, known or presumed infecting organism and susceptibility, and the patient's tolerance.⁹ When presenting symptoms are mild, it is preferable to delay initiation of antimicrobial therapy until results of the urine culture are available. Patients presenting to the emergency department with UTI without signs of septic shock do not have worse outcomes if antibiotic therapy is not given immediately.¹⁵⁴

For severe symptoms, empirical antimicrobial therapy is initiated pending urine culture results. Previous urine culture results, if available, and recent antimicrobial therapy received by the patient should be considered in the selection of the empiric regimen.⁹ Parenteral therapy is indicated for patients with hemodynamic instability, patients who cannot tolerate or absorb oral medications, or who are known or suspected to have an infecting organism resistant to available oral options. Patients with sepsis including septic shock should receive initial empirical broad-spectrum antimicrobial therapy.¹⁴⁷ Appropriate regimens for these patients may include an aminoglycoside, with or without ampicillin; a cephalosporin with an aminoglycoside; a carbapenem (imipenem, meropenem); an extended-spectrum cephalosporin, or a β -lactam/ β -lactamase inhibitor. The β -lactam/ β -lactamase inhibitors—ceftazidime/avibactam and ceftolozane/tazobactam—are effective for the treatment of complicated infection with resistant organisms.¹⁵⁵⁻¹⁵⁹ Intravenous fosfomycin has also been used for some highly resistant organisms; given potential for resistance development, it should be used in combination with another broad-spectrum antimicrobial.¹⁶⁰

Fluoroquinolones with good urinary excretion and broad gram-negative coverage—norfloxacin, ciprofloxacin, and levofloxacin—are indicated for empirical oral therapy⁹; norfloxacin is useful for catheter-related bladder infection but is not effective for renal or prostate infection. Many other oral agents are effective and may be appropriate, depending on patient tolerance and the specific infecting organism.⁹ These include TMP/SMX, amoxicillin, amoxicillin/clavulanic acid, oral cephalosporins, and doxycycline. Nitrofurantoin may be effective for treatment of catheter-related bladder infection, but it is not effective for renal or prostate infection and is contraindicated for

treatment of patients with renal failure because peripheral neuropathy has been reported to occur with accumulation of toxic metabolites.

Substantial clinical improvement is expected by 48 to 72 hours after initiation of effective antimicrobial therapy. Empirical therapy is then reassessed according to culture results, with switch to a more narrow-spectrum agent. Treatment courses of 5 to 7 days are effective for both catheter-related UTI confined to the lower urinary tract and obstructive pyelonephritis, where the obstruction has been removed and the patient has responded to initial therapy.¹⁶¹

OTHER INTERVENTIONS

Appropriate supportive care must be initiated promptly. The removal and replacement of a long-term indwelling catheter before institution of antimicrobial therapy are associated with a more rapid defervescence and a lower risk of early relapse after therapy, as well as facilitating collection of a more valid urine culture specimen.³¹ The clinical benefits of catheter replacement are presumed to result from removal of the high concentration of organisms in the biofilm, which are often not eradicated by antimicrobial therapy and remain a source for relapse. Urologic investigations such as cystoscopy, retrograde pyelography, and urodynamic studies should be obtained as appropriate. If encrusted *C. urealyticum* infection is present, surgical resection of the encrustations is required together with antimicrobial therapy. *C. urealyticum* strains are generally susceptible to vancomycin, tetracyclines, and fluoroquinolones.¹⁴²

MANAGEMENT OF RECURRENT INFECTION

When symptoms recur early after appropriate antimicrobial treatment for a susceptible organism, further evaluation is necessary to identify abnormalities such as abscesses. Indwelling devices should be removed whenever possible; duration of catheter use is the most important risk factor for UTI.¹⁶² Evidence-based infection control guidelines provide recommendations for prevention of catheter-acquired UTI,¹⁶² and multifaceted programs are effective in decreasing catheter use and infection rates.¹⁶³ Specific practices include catheter use for restricted indications, limiting duration of use, safe practices for insertion and care, and appropriate catheter size. Interventions not shown to be effective and not recommended are use of different catheter materials, antimicrobial coating of catheters, antiseptic or antimicrobial meatal care, and instillation of antiseptics into the drainage bag. Patients with long-term indwelling catheters and other indwelling devices experience infection because of biofilm formation on these devices; the prevention of catheter-acquired infection will ultimately require development of biofilm-resistant biomaterials.^{164,165}

Long-term prophylactic antimicrobial therapy is not recommended.^{9,132,166} When patients with impaired voiding or indwelling devices are given prophylactic antimicrobial agents, there is little, if any, decrease in symptomatic infection, and rapid reinfection with resistant organisms is uniformly observed.⁹ For selected patients with a persistent abnormality and symptomatic relapsing infection with an organism that cannot be eradicated, suppressive therapy may be considered. Antimicrobial options are limited, however, given rapid emergence of resistance; nitrofurantoin should

be favored where possible given low rates of emergence of multidrug resistance despite its long-term use,¹⁶⁷ and the patient and general practitioner should be reminded that nitrofurantoin suffices to sterilize the urine but does not treat active soft tissue infection. If the patient remains clinically stable and the urine is sterile, this dose is usually decreased to about half (for nitrofurantoin, from 100 mg qd to 50 mg qd) after 4 to 6 weeks. Suppressive therapy is not recommended for patients with indwelling devices because biofilm formation facilitates rapid reinfection and emergence of resistant organisms. However, short-term use of such therapy (several weeks or months) for patients with complex urologic abnormalities and indwelling devices may occasionally be considered as part of a palliative care strategy.

Novel nonantibiotic approaches are being attempted for UTI due to impaired bladder emptying and other underlying abnormalities. “Bacterial interference,” in which asymptomatic bacteriuria with a nonpathogenic *E. coli* strain is established to prevent infection by other more virulent strains, has been attempted yet without long-term success.¹⁶⁸ There is currently avid interest in bacteriophage therapy, though most data stem from individual case reports with little long-term follow-up, where phages were often given in addition to antibiotics. In one randomized trial of men with UTI, a standard phage cocktail (not individualized to participants’ infecting strains) instilled into the bladder led to lower rates of microbiologic cure than placebo or antibiotic therapy.¹⁶⁹

COMPLICATED UTI: PROSTATITIS

The development of the National Institutes of Health classification of prostatitis syndromes (Table 38.6) and its subsequent application in clinical trials and care have greatly advanced the understanding and appropriate management of this common problem.¹⁷⁰ Only acute or chronic bacterial prostatitis is considered attributable to infection and has indications for antimicrobial therapy.

ACUTE BACTERIAL PROSTATITIS

Acute bacterial prostatitis is a severe infection and a urologic emergency.¹⁷¹ It is usually community acquired, although infection may occur after prostate biopsy.¹⁷² Affected men present with severe systemic manifestations including fever and marked urinary symptoms of dysuria and frequency. Urinary obstruction and intense suprapubic pain are often present. Bacteremia is reported in roughly a quarter of episodes.

E. coli is isolated in approximately 70% of episodes. These strains are characterized by the presence of multiple virulence factors.¹⁷³ *Proteus* species, *Klebsiella* species, *P. aeruginosa*, *S. aureus*, and *Enterococcus* species are each isolated in less than 10% of cases.¹⁷²

Diagnosis is clinical and is supported by physical examination, laboratory studies, and occasionally imaging findings. Although there may be theoretical concern that digital rectal examination could cause bacteremia in acute prostatitis due to a high bacterial load in the prostate, there is no clinical evidence confirming the association. While extensive prostate massage should be avoided, a brief, gentle examination to identify acute tenderness or

Table 38.6 National Institutes of Health Classification of Prostatitis Syndromes

Class	Description
I	Acute bacterial prostatitis
II	Chronic bacterial prostatitis
III	Chronic pelvic pain syndrome (CPPS)
IIIa	Inflammatory CPPS Leukocytes present in semen, in urine after prostate massage, or in expressed prostate secretions
IIIb	Noninflammatory CPPS Absence of leukocytes in specimens
IV	Asymptomatic inflammatory prostatitis Leukocytes in specimens similar to those in inflammatory CPPS, but no symptoms

Data from Krieger JN, Nyberg L Jr, Nickel JC. NIH consensus definition and classification of prostatitis. *JAMA*. 1999;282(3):236–237.

any gross abnormalities such as fluctuance may be key to distinguishing acute prostatitis from pyelonephritis in men.

There is no gold standard for the laboratory diagnosis of acute bacterial prostatitis, but urinalysis and urine culture are important for confirming clinical suspicion and guiding antibiotic therapy, respectively. Blood cultures are not universally required, and in one study, they contributed to microbiologic diagnosis in only 5% of patients.¹⁷⁴ Yet blood culture collection is justified given the possible diagnostic uncertainty early in the course of the disease, as well as the clinical reality that urine may be collected only after the first dose of antibiotic has been given, reducing diagnostic sensitivity.¹⁷⁵ Prostate-specific antigen has not shown sufficient sensitivity or specificity to be used as a diagnostic tool. Imaging is not required for initial diagnosis of acute prostatitis. Both transrectal ultrasound and MRI can be useful for identification of prostatic abscess; however, routine evaluation for abscesses is not recommended, as studies have shown low (0%–2.7%) rates.¹⁷⁴

Management includes bladder drainage by insertion of a urethral or suprapubic catheter and initiation of empirical parenteral antimicrobial therapy. Most antimicrobial agents are active in the acutely inflamed prostate. A combination of a β -lactam and an aminoglycoside is still considered first-line therapy, although other broad-spectrum parenteral antibiotics, such as piperacillin/tazobactam and carbapenems, are also effective.¹⁷⁶ Once the infecting organism and susceptibility are confirmed and there has been an adequate clinical response to parenteral therapy, the antibiotic is modified to oral therapy to complete a 2- to 4-week course. These durations are based on expert opinion, but evidence from a recent randomized trial suggests that 2 weeks should be the minimum duration. The trial compared 1 versus 2 weeks of ofloxacin therapy in acute prostatitis due to susceptible uropathogens and failed to show noninferiority, with 56% and 78% experiencing treatment success, respectively, at 6 weeks.¹⁷⁷

A fluoroquinolone, either ciprofloxacin or levofloxacin, is recommended as oral therapy if the infecting organism is susceptible. When there is not a prompt clinical response

following bladder drainage and initiation of effective antimicrobial therapy, CT or MRI is indicated to search for the uncommon complication of prostate abscess. When an abscess is identified, transrectal ultrasonography-guided aspiration is usually effective for drainage. A small proportion of patients, 10% to 15%, develop chronic bacterial prostatitis following acute prostatitis.¹⁷³

CHRONIC BACTERIAL PROSTATITIS

Chronic bacterial prostatitis occurs in men with persistent prostate infection.^{171,178–181} Bacteria enter the prostate from the urethra and persist because of limited antimicrobial diffusion or activity in the gland, as well as the frequent presence of infected prostate stones in older men. A common clinical presentation is recurrent acute cystitis because bacteria in the prostate intermittently enter the bladder. The same organism is often repeatedly isolated, but the intervals between symptomatic episodes may last months or even years. Other symptoms are generally mild, such as irritative voiding symptoms and discomfort localized to the testicles, lower back, or perineum. Results of the prostate examination are usually normal, but tenderness may occasionally be elicited.

Chronic bacterial prostatitis is documented microbiologically in only 10% of men who present with the clinical syndrome of chronic prostatitis or chronic pelvic pain syndrome (CPPS). Gram-negative organisms including *Enterobacteriaceae* and *P. aeruginosa* are the bacteria most commonly isolated; gram-positive *Enterococcus* species, *S. aureus*, coagulase-negative staphylococci, and group B streptococci may also be isolated, though their presence can be confusing because it may also reflect culture contamination by fecal or skin flora.¹⁷³ Sexually transmitted organisms such as *Chlamydia trachomatis*, *U. urealyticum*, *Mycoplasma genitalium*, and *Trichomonas vaginalis* are uncommon and, when present, are usually identified in younger men.¹⁷⁶

Confirmation of the diagnosis requires the two-glass test, in which a midstream urine sample is collected for urinalysis and culture, prostate massage is then performed for roughly a minute, and immediately thereafter first-stream urine is collected for urinalysis and culture. In a positive test, the postmassage sample will reveal more pyuria and at least one log more bacterial colony-forming units than the premassage specimen.¹⁷⁶

The test's sensitivity and specificity are wanting, however. In one study of 463 patients and 121 age-matched controls, 70% of patients were found to harbor at least one organism in a postprostatic massage specimen, and uropathogens such as *E. coli* were isolated from 8% of patients and 8.3% of controls.¹⁸² In another study, gram-positive organisms were isolated from 6% of 470 men after prostatic massage, but 97% of these organisms were not confirmed on second culture, suggesting contamination or colonization.¹⁸³

Despite this uncertainty, chronic bacterial prostatitis does respond to appropriate antimicrobial treatment, although relapse after treatment is common. Ciprofloxacin is the first choice for antimicrobial treatment of chronic bacterial prostatitis when susceptible organisms are isolated. These agents penetrate well into the prostate and seminal fluid, and they remain active in the acidic environment of the prostate. Cure rates at 6 months after a 4-week course are

75% to 89%,^{176,178} although late relapses may still occur. Trimethoprim/sulfamethoxazole (TMP-SMX) is often the first alternative therapy used: It appears to have excellent prostatic penetration, particularly the trimethoprim component.¹⁷⁶ Data supporting its therapeutic use are sparse, however, coming mainly from older studies with smaller numbers of patients and longer (90-day) treatment courses. Doxycycline and macrolides are considered second-line drugs but are preferred for gram-positive infections.¹⁷⁸ Men who present for the first time with CPPS and with evidence of inflammation (i.e., leukocytes) in expressed prostatic secretions, but in whom cultures are negative, should be prescribed a 4-week trial of antimicrobial therapy if they have not previously received a prolonged antimicrobial course.^{178,184} In reported case series, up to 10% of men with such findings show response to antimicrobial therapy despite negative culture results; however, comparative clinical trials in treatment-naïve men with negative culture results have not been reported. If symptoms persist or recur after this 4-week antimicrobial trial, and if results of postprostatic massage cultures remain negative, further antimicrobial therapy is not indicated and patients may be referred for perineal physical therapy.^{179,185,186}

ASYMPTOMATIC BACTERIURIA

Asymptomatic bacteriuria (ASB) is a common finding, particularly in women, older persons, and some patients with persistent genitourinary abnormalities, that does not require antibiotic therapy. The three clinical scenarios traditionally considered by some to be exceptions to this rule have been pregnancy, recent (≤ 3 months) renal transplantation, and impending traumatic urologic procedures. Yet the true incidence and risks of pyelonephritis in pregnancy may be much lower than those reported by early studies, leading some to question the need for universal screening and treatment of ASB in this population, and screening and treatment of ASB after renal transplantation have never been shown to improve clinical outcomes of transplant recipients. An extensive guideline has been updated with a review of ASB and the evidence, and lack of evidence, for its treatment.¹⁴⁶

EPIDEMIOLOGY

The prevalence of bacteriuria among sexually active young women ranges from 3% to 5% but is $<1\%$ among age-matched controls who are not sexually active. Bacteriuria is present in 5% to 10% of healthy postmenopausal women¹⁸⁷ and in 20% of women older than 80 years living in the community.¹⁸⁸ Asymptomatic bacteriuria is uncommon in younger men, but the prevalence increases in men older than 65 years, presumably concurrently with age-related prostate hypertrophy. Bacteriuria occurs in 10% of healthy men older than 80 years living in the community.¹⁸⁸ Among residents of nursing homes who do not have indwelling catheters, 20% to 50% of women and 15% to 40% of men are bacteriuric. The prevalence among persons with long-term indwelling catheters is 100%.¹⁴⁹ Among spinal cord-injured patients with impaired voiding and no indwelling catheter, the prevalence of bacteriuria is 50%, regardless of the method used for bladder emptying.¹³²

Patients with an increased prevalence of ASB also have a higher incidence of symptomatic UTI—yet this higher frequency is not attributable to bacteriuria. The same biological determinants promote both asymptomatic and symptomatic infection. Bacteriuria in healthy young women is usually transient, but up to 8% have acute cystitis within 1 week of initial identification of a positive urine culture result.¹⁸⁹ In women with diabetes¹⁹⁰ and female or male residents of nursing homes,¹⁴⁹ persistent bacteriuria for months or years, frequently with the same strain, is often observed. No long-term negative outcomes have been attributed to asymptomatic bacteriuria.^{187,190} Bacteriuric individuals are not at increased risk of hypertension or chronic renal failure, and survival is similar to that for persons without bacteriuria.

Harmful short-term outcomes attributable to asymptomatic bacteriuria have been recognized in only two distinct populations: pregnant women and patients who undergo traumatic genitourinary procedures.¹⁸⁷ The prevalence of bacteriuria during early pregnancy is 3% to 7%, similar to that among age-matched nonpregnant women. The physiologic changes that accompany the increased progesterone levels in pregnancy include smooth muscle relaxation and decreased peristalsis, which result in dilation of the renal pelvis and ureters. In later pregnancy, ureteric obstruction may result from pressure of the uterus at the pelvic brim. Earlier studies reported that in 20% to 35% of pregnant women with untreated bacteriuria, acute pyelonephritis developed later in pregnancy, usually at the end of the second trimester or early in the third trimester; this incidence of pyelonephritis was 20-fold to 30-fold higher among untreated women with bacteriuria than among women whose initial screening urine cultures yielded negative results or in whom bacteriuria was treated. In these earlier studies, acute pyelonephritis in later pregnancy was associated with premature delivery and poorer fetal outcomes.

A 2015 randomized trial from the Netherlands, however, where screening for ASB in pregnancy is not part of routine care, calls into question the incidence and outcomes reported in earlier studies.¹⁹¹ While pyelonephritis occurred more frequently in pregnant participants with untreated ASB than in those without ASB or with treated ASB, it occurred in only 2.4% of women—much less than the 20% to 35% reported in earlier studies—and its occurrence did not lead to adverse fetal outcomes.

The second group of bacteriuric patients at risk are those who undergo traumatic urologic procedures. If bacteriuria remains untreated, bacteremia develops in up to 60% after the procedure and in 5% to 10% progresses to severe sepsis or septic shock.

PATHOGENESIS AND MICROBIOLOGY

E. coli is isolated from 80% of healthy women with asymptomatic bacteriuria.¹⁸⁹ Most of the remaining bacterial strains are reported to be *K. pneumoniae*, *Enterococcus* species, and coagulase-negative staphylococci, but isolation of the latter two may be a result of urine sample contamination from the distal urethra.³⁴ For men older than 65 years of age, coagulase-negative staphylococci are isolated most frequently, followed by *E. coli* and *Enterococcus* species. In patients with underlying genitourinary abnormalities, a

wider variety of organisms are isolated. *E. coli* and other organisms associated with ASB are characterized by the relative absence of or lack of expression of recognized virulence factors.¹⁹² Some strains that are strongly adherent but unable to stimulate an epithelial cell IL-6 cytokine response have been described. These relatively avirulent *E. coli* strains may originate from nonvirulent commensal strains or evolve from virulent strains by attenuation of virulence genes.^{193,194} Other organisms frequently isolated, such as coagulase-negative staphylococci and *Enterococcus* species, are relatively nonpathogenic and seldom associated with symptomatic infection. Biofilm formation results in polymicrobial bacteriuria in patients with chronic indwelling catheters.¹³⁷

HOST FACTORS

The genetic and behavioral risk factors and genitourinary abnormalities associated with ASB are similar to those described for uncomplicated and complicated symptomatic UTI. For younger women, behavioral risks are sexual activity and spermicide use.¹⁸⁹ Bacteriuric women older than 80 years who reside in the community are characterized by reduced mobility, urinary incontinence, and receiving estrogen treatment; and men older than 80 years with bacteriuria are characterized by prostate disease, history of stroke, and residence in supervised housing.¹⁸⁸ Functional impairment is the major risk factor associated with asymptomatic bacteriuria in the long-term care facility population without indwelling catheters.¹⁴⁹ Higher residual urine volume is not associated with increased bacteriuria in elderly populations but has been associated with bacteriuria in patients referred to an ambulatory urology clinic.¹⁴⁹

DIAGNOSIS

The criterion for identification of ASB is an organism quantitative count of $\geq 10^5$ CFU/mL in a urine culture. For women, two consecutive urine specimens with similar culture results are recommended, but a single specimen is sufficient for men.¹⁸⁷ When an initial urine culture in a young woman yields positive results, a second positive result in a specimen obtained within 2 weeks confirms bacteriuria in 85% to 90% of women. If the second specimen result is negative, the initial positive culture result may have been contaminated; however, bacteriuria may also have resolved spontaneously. If a voided specimen has a low quantitative count of a single potential uropathogen, and if it is essential to rule out bacteriuria, a second urine specimen should be collected as a first morning void.

Pyuria usually accompanies bacteriuria, but there is variability in the frequency of pyuria observed in different populations. Pyuria is present in only 50% of bacteriuric pregnant women; therefore screening for pyuria is not a reliable method to rule out bacteriuria in pregnancy.¹⁹⁵ Pyuria is present in about 75% of diabetic women with bacteriuria, in 90% of bacteriuric patients undergoing hemodialysis, and in >90% of elderly persons with bacteriuria.¹⁸⁷ Pyuria also occurs in 30% to 70% of bacteriuric patients with short-term indwelling catheters and in 100% of those with long-term indwelling catheters. Some other biomarkers have been explored to be used in differentiating asymptomatic from symptomatic infection, particularly in the nursing home population.

Urinary IL-6 has some association with symptomatic infection but has not been shown, to date, to have clinical utility.^{196,197}

TREATMENT

Screening for and treatment of ASB is recommended only for pregnant women or for patients who will undergo a traumatic genitourinary tract procedure.¹⁴⁶ For other patients, treatment does not improve short- or long-term clinical outcomes, whereas negative consequences such as reinfection with organisms of increased antimicrobial resistance and adverse drug effects occur.^{187,198} In girls and women, an increased frequency of symptomatic infection has been reported after treatment of asymptomatic bacteriuria.^{187,199} This finding may be attributable to alteration of vaginal flora by antimicrobial therapy or replacement of a benign organism in the urine by a more virulent organism. Studies suggest no benefits of treatment of asymptomatic bacteriuria for patients with complicated infection including renal transplant recipients.²⁰⁰⁻²⁰² Treatment of patients with asymptomatic bacteriuria is identified as a major contributor to inappropriate antimicrobial use and is an important target for intervention in antimicrobial stewardship programs.^{203,204}

Current recommendations are that pregnant women be screened for bacteriuria by urine culture at the end of the first trimester and treated once (without follow-up surveillance cultures) if bacteriuria is found.¹⁴⁶ Preferred regimens include a 5- or 7-day course of nitrofurantoin²⁰⁵ or a 7-day course of amoxicillin, amoxicillin/clavulanic acid, or a cephalosporin. The specific regimen is chosen on the basis of the organism isolated and the patient's tolerance. Shorter antimicrobial courses are sometimes prescribed, but these abbreviated courses are likely less effective.²⁰⁶ TMP/SMX should be avoided, especially in the first trimester, and fluoroquinolones are contraindicated during pregnancy.

Initiation of effective antimicrobial treatment immediately before a traumatic urologic procedure prevents bacteremia and sepsis in a bacteriuric patient.^{187,207} Conceptually, this approach is surgical prophylaxis rather than treatment of asymptomatic bacteriuria. A single dose of an antimicrobial agent is usually adequate, although some guidelines recommend that the antimicrobial agent be continued after transurethral resection of the prostate until the indwelling catheter is removed.¹⁸⁷ Antimicrobial agents are not recommended for minor urologic procedures, such as cystoscopy and urodynamic studies or before replacement of a long-term urethral catheter, as the risk of bacteremia and sepsis with these interventions is low and clinical outcomes are not improved with prophylactic antimicrobial agents.^{9,208,209}

URINARY TRACT INFECTION IN DIFFERENT PATIENT POPULATIONS

URINARY TRACT INFECTION IN CHILDREN

While this chapter focuses on UTI in adults, infections are common in children. A brief description of similarities and differences of UTI in these two populations is provided in [Box 38.1](#).

Box 38.1 Comparison of Urinary Tract Infection (UTI) Between Adults and Children**Similarities in UTI Between Adults and Children**

- UTIs are common in childhood, with an incidence of cystitis between 8.4–40.3 per 1000 person-years in girls and 2.6–5.2 in boys and an incidence of pyelonephritis between 1.4–5.5 per 1000 person-years in girls and 0.1–5.4 per 1000 person-years in boys, varying with age.²¹⁰ UTI is the most common definitive bacterial infection (i.e., confirmed by culture, among children and young persons [CYPs]).
- UTI is classified according to site (primarily cystitis vs. pyelonephritis), episode (first/sporadic vs. recurrent vs. breakthrough in patients receiving antimicrobial prophylaxis), symptoms (asymptomatic vs. symptomatic), and complicating factors; the approach to classification, however, may differ between CYP and adults.²¹¹
- To exclude or confirm the presence of a UTI, urine must be sampled before any antibiotic treatment. In toilet-trained CYP, suitable samples for microscopy and culture are obtained using the same methods as in adults.
- The microbiologic epidemiology of UTI in CYP is similar to that in adults, with *E. coli* being the leading causative pathogen. Identification of other bacteria on culture, especially *Klebsiella* spp. and *Pseudomonas* spp., should alert the clinician to the potential presence of an underlying congenital abnormality of the kidney or urinary tract (CAKUT).²¹¹

Differences in UTI Between Adults and Children

- Cystitis can be difficult to identify in young non-toilet-trained children unable to report typical symptoms. Similarly, pyelonephritis may present with nonspecific symptoms and signs, such as malaise and vomiting, together with fever, but without localizing signs, such as flank pain.²¹¹
- While cystitis more commonly affects female CYPs of all ages, in the first year of life the incidence of pyelonephritis is comparable for males (5.4 per 1000 person-years) and females (4.9 per 1000 person-years).²¹⁰
- Febrile UTI (pyelonephritis) is often the first manifestation of CAKUT. Around 4 in 10 children with UTI younger than age 2 who

are investigated using abdominal ultrasound are found to have unilateral or bilateral hydronephrosis or anatomic evidence of CAKUT.²¹² Renal and bladder ultrasound are therefore recommended after any first febrile UTI in childhood, and additional radiologic evaluations may be necessary in CYP with recurrent UTI or abnormal sonographic findings.²¹¹ CAKUT affects males almost 1.5 times as often as females.²¹³ UTIs in CYP with CAKUT are considered complicated because they occur on a background of a known mechanical or functional pathology of the urinary tract.²¹¹

- Typical risk factors for UTI in adults, in particular urinary catheters, type 2 diabetes, and diseases primarily or secondarily associated with a state of immunosuppression, are rare among CYP. Instead, bladder-bowel dysfunction, vesicoureteral reflux, and other CAKUT are common risk factors.²¹¹
- Because UTI, especially pyelonephritis, more commonly affects younger children who may not yet be toilet trained, urine samples are typically obtained by one of the following four methods: urine bag attached to cleaned genitalia, spontaneous voiding into a sterile container (called clean-catch urine), transurethral bladder catheterization, or suprapubic bladder aspiration. The methods are listed in order from that most to that least prone to contamination. Bag and clean-catch urine samples are most helpful to rule out UTI.²¹⁴
- In young children, UTI occurs in a urinary tract still undergoing developmental changes, thus potentially more prone to remodeling in response to repeated bouts of infection and inflammation.²¹⁵ This is thought to be the reason that recurrent UTIs in childhood are associated with renal scarring, which is, in turn, associated with impaired renal function in adulthood.^{216,217} For this reason, there is ongoing debate regarding the benefits of long-term antibiotic prophylaxis in at-risk children with CAKUT. While antibiotic prophylaxis may offer some protection against recurrent symptomatic UTI, the benefits are likely small²¹⁸ and must be balanced against a considerably increased risk of further episodes being caused by resistant, difficult-to-treat pathogens.²¹⁹

RENAL TRANSPLANT RECIPIENTS

UTI accounts for 45% to 69% of infections that occur in patients after renal transplantation.^{220,221} By 6 months after transplantation, 17% of recipients have experienced at least one UTI, and by 3 years, infection is reported in 60% of female recipients and 47% of male recipients.²²² The incidence is highest in the early posttransplant period, when immunosuppressive therapy is most intense and infection complicates surgical intervention and the use of urologic devices, such as urinary catheters and ureteric stents.^{223–225} Asymptomatic bacteriuria, cystitis, and pyelonephritis are common. Early posttransplantation infections tend to be more severe and to manifest more frequently as pyelonephritis or with bacteremia.²²⁶

Risk factors for posttransplantation UTI are 1. patient-specific, such as female gender, older age, diabetes mellitus, pretransplantation UTI, prolonged prior dialysis, and polycystic kidney disease; or 2. transplant procedure related, such as allograft trauma, microbial contamination of cadaveric kidneys, technical complications related to ureteral anastomosis, the presence of urinary catheters and ureteric stents, immunosuppression, reimplantation,

and vesicoureteric reflux.²²⁵ Independent risk factors for asymptomatic bacteriuria and for infection in the first year after transplantation are older age, female gender, higher number of days of indwelling catheter after transplant, anatomic genitourinary abnormalities, and urinary infection within 1 month before the procedure.²²⁷ For acute pyelonephritis at any time after transplantation including beyond the first year, independent risk factors are female gender, experiencing acute rejection episodes, higher number of UTI, and receipt of mycophenolate mofetil.²²⁸ Following transplantation, the risk for UTI correlates with the duration of perioperative catheterization or ureteral stent.^{229,230} The routine use of ureteric stents at transplantation decreases the overall transplantation complication rate, but stent implantation is associated with a small but significant increased risk of UTI.

While asymptomatic bacteriuria has some association with symptomatic UTI, it is not a risk factor for graft loss or impaired graft function.¹⁸⁷ Two randomized trials have shown that screening and treating asymptomatic bacteriuria after kidney transplantation does not improve clinical outcomes of recipients. Conversely, this strategy increases

antibiotic use and promotes emergence of more resistant organisms in the urine.^{231,232} Transplant recipients with asymptomatic bacteriuria who progress to graft failure usually also experience symptomatic infection. Graft failure in these patients is usually attributable to urologic abnormalities that promote infection rather than a direct consequence of infection. Symptomatic UTI has also not been independently associated with graft survival in the majority of large studies.^{223,224,233-235} However, other small studies and case reports have described transplant recipients receiving stable immunosuppressive regimens who experience deterioration in graft function that is coincident with an episode of acute pyelonephritis.²³⁶⁻²³⁸ This occurrence may be attributable to activation of the immune system by the infection and ensuing inflammation in patients with one functioning kidney.

The principles of management are similar to those for any patient with complicated UTI. They include prompt clinical diagnosis and initiation of antimicrobial therapy, obtaining appropriate specimens for culture and urologic evaluation for underlying genitourinary abnormalities that may promote infection. Enterobacteriaceae, particularly *E. coli*, are the most common infecting organisms, but a variety of other bacteria or yeasts may be isolated. The choice of antimicrobial agent is determined by the susceptibility of the infecting organism and the patient's tolerance. A 1-week course of antimicrobial agent for cystitis and 2 weeks for pyelonephritis is recommended. The effect of type and intensity of immunosuppressive therapy on antimicrobial efficacy has not been reported. Bacteriuria does not compromise renal function,^{202,234} and treatment of bacteriuria does not improve clinical outcomes including the frequency of symptomatic episodes or graft function.²⁰⁰⁻²⁰² Current guidelines therefore do not recommend screening for bacteriuria.^{146,239} Treatment of asymptomatic candiduria in renal transplant recipients has also not been shown to be beneficial and is not recommended.²⁴⁰

UTIs after transplantation are limited by ensuring optimal surgical technique, which includes minimizing the perioperative duration of indwelling urinary devices. TMP/SMX prophylaxis given in the posttransplant period (usually for pneumocystis prophylaxis) decreases the risk of UTI and ASB in the short term, but effectiveness is compromised by increasing antimicrobial resistance.^{241,242} When UTI is diagnosed in patients receiving TMP/SMX prophylaxis, isolated organisms are invariably resistant to TMP/SMX. Frequent recurrent symptomatic infection after transplantation may be attributable to an inadequate duration of antimicrobial treatment, resistance of the infecting organism, the presence of urologic abnormalities such as obstruction and stones, or infection of native kidneys. When infection recurs shortly after treatment, previous urine culture results should be reviewed to establish whether reinfection or relapse is occurring and to confirm that the infecting organism is susceptible to the antimicrobial therapy given. Repeated relapse after prolonged antimicrobial therapy in patients without identified urologic abnormalities is often caused by infection localized to the native kidneys. Such patients usually have a history of recurrent UTI before transplantation. The organisms frequently cannot be eradicated, presumably because of failure of antibiotics

to achieve effective levels in the nonfunctioning kidney. Removal of the native kidney, especially in the case of abscess, may be necessary to prevent further symptomatic episodes in these patients.

PERSONS WITH RENAL FAILURE

Infections in patients with mild-to-moderate renal failure usually respond adequately to antimicrobial therapy.²⁴³ When renal impairment is severe, however, adequate urine or renal levels of antimicrobial agents may not be achieved because of limited kidney perfusion. If renal failure occurs with acute pyelonephritis, the response following initiation of antimicrobial therapy may be delayed or the risk of relapse after therapy may be increased. Aminoglycosides have little penetration into nonfunctioning kidneys and are not recommended for treatment. Fluoroquinolones, TMP/SMX, ampicillin, and cephalosporins have all been effective for treatment in individual case reports. Ertapenem therapy for patients with cystitis or catheter-acquired UTI is effective but associated with a more prolonged time to negative cultures for patients with impaired renal function.²⁴⁴ In patients infected with multidrug-resistant organisms treated with fosfomycin, chronic kidney disease was associated with a significantly increased risk of treatment failure.²⁴⁵

Another therapeutic problem arises when infection is localized to a unilateral nonfunctioning kidney. Antimicrobial therapy sterilizes the urine and often ameliorates symptoms because high urine levels are achieved by antimicrobial excretion through the functioning kidney. However, organisms persist in the kidney with impaired function because antimicrobial penetration to the site of infection is inadequate. When antimicrobial therapy is discontinued, a prompt relapse of infection occurs. Localization of infection to one kidney can be documented by culture of urine collected directly from the ureter following bladder irrigation with normal saline to remove infected bladder urine before ureteric catheterization. For symptomatic relapsing infection when a nonfunctioning kidney is known or suspected to be infected, management options include a trial of a more prolonged course of antimicrobial therapy, continuous suppressive therapy, and surgical removal of the nonfunctioning kidney.

PERSONS WITH URINARY STONES

UTI complicates urolithiasis in several ways; infection may be the cause of stone formation, noninfected metabolic stones may become colonized with bacteria that persist in biofilm, and obstructing noninfected stones may precipitate infection proximal to the obstruction.²⁴⁶ In any patient with urolithiasis and infection, the infection should be controlled with appropriate antimicrobial therapy before urologic manipulations.²⁴⁷

Infection stones, also called "struvite stones," are a complication of infection with urease-producing organisms such as *P. mirabilis*.²⁴⁸ Urease catalyzes the hydrolysis of urea in the urine. This process produces ammonia and alkaline urine, favoring precipitation of magnesium ammonium phosphate, carbonate apatite, and monoammonium urate.²⁴⁹ These crystals are incorporated into bacterial

biofilm, creating the infection stone.¹³⁷ Infection stones continue to enlarge, sometimes rapidly, and ultimately cause obstruction and renal failure if not adequately treated. Patients at increased risk for development of infection stones are those with long-term indwelling catheters, urinary tract obstruction, neurogenic voiding dysfunction, distal renal tubular acidosis, and medullary sponge kidney (see also Chapter 40).

Management of infection stones requires complete removal of the stone, together with sterilization of the urine.²⁴⁷ Antimicrobial therapy, selected on the basis of urine culture results, is continued until all fragments are passed. The optimal duration of antimicrobial therapy after lithotripsy is unknown, but in cases where stone removal is successful and source control has thus been achieved, a short course (≤ 5 days) is reasonable. When percutaneous lithotripsy is not effective or is contraindicated, open surgery for stone removal or nephrectomy may be necessary. For selected elderly or debilitated patients with complex urologic or medical problems, stone removal may not be possible. In such patients, continuous suppressive antimicrobial therapy is recommended to limit stone enlargement and preserve renal function.²⁵⁰

Bacteria may adhere to the surface of a stone that is not initially infected and may subsequently persist in biofilm.¹³⁷ Culture of a voided urine specimen in such cases often does not reflect the bacteriology of the stone. In a series of 75 patients with renal stones that were not infection stones, 36 (49%) of the stones were colonized but only 19 (53%) of these patients also had bacteriuria.²⁵¹ Organisms isolated from the colonized stones included *E. coli* (75%), *Enterococcus* species (100%), *P. aeruginosa* (19%), *Klebsiella* species (31%), *P. mirabilis* (8.3%), *Streptococcus* species (31%), *Citrobacter* species (8.3%), *S. saprophyticus* (19%), *M. morganii* (8.3%), and *Gemella* species (8.3%). Larger stones were more likely to be colonized. Patients with colonized stones are at increased risk for postprocedure sepsis, even when the voided urine specimen culture has negative results.²⁵² Perioperative antimicrobial agents are recommended for all patients with stones who have indwelling catheters or stents because they are at increased risk of stone colonization. Even with antimicrobial prophylaxis, 10% of patients undergoing percutaneous nephrolithotomy experience postoperative fever. Positive urine culture, diabetes, staghorn calculi, and preoperative nephrostomy are risk factors for postprocedure fever.²⁵³

A noninfected stone causing ureteric obstruction may be complicated by infection of undrained urine proximal to the stone. Complete obstruction is associated with a high risk for bacteremia and sepsis, and prompt drainage is essential for these patients.²⁵⁴ When a ureter is obstructed, the microbiology of the voided urine specimen may not be consistent with that of urine sampled directly from the renal pelvis. Among patients with obstruction and a positive culture for urine sampled from the renal pelvis, the voided urine specimen culture had positive results in only 16% of cases in one report, and the organisms isolated were concordant between the two specimens in only 23% of patients.²⁵⁵ Thus a urine specimen should be obtained for culture by percutaneous aspiration of the renal pelvis in patients with obstruction, whenever possible.

OTHER INFECTIONS OF THE URINARY TRACT

RENAL AND PERINEPHRIC ABSCESES

Renal and perinephric abscesses are uncommon suppurative complications associated with substantial morbidity and mortality. Renal abscesses are located entirely within the renal parenchyma, whereas perinephric abscesses occupy the retroperitoneal fat and fascia surrounding the kidney. Abscesses may involve both the renal and perinephric tissues: 25% to 39% of abscesses are intranephric, 19% to 25% are both intranephric and perinephric, and 42% to 51% are perinephric alone.^{256,257} These abscesses develop following ascending UTI and pyelonephritis or after hematogenous spread to the renal cortex or retroperitoneum following bacteremia from another site. Multiple bilateral cortical microabscesses suggest hematogenous spread. Complicating factors such as diabetes, urolithiasis, and obstruction are present in most cases. A population-based study from Taiwan reported an incidence of hospitalization for renal and perinephric abscess of 4.6/10,000 person-years for diabetic patients and 1.1/10,000 for nondiabetic patients.²⁵⁸

The organisms most commonly isolated from abscesses are *E. coli*, *K. pneumoniae*, *P. mirabilis*, *S. aureus*, and anaerobes. *S. aureus* abscesses are most likely to have originated with hematogenous spread from another site of infection. Clinical manifestations mimic those of acute pyelonephritis. Clinical course in patients treated initially for pyelonephritis is characterized by delayed clinical response or early symptomatic relapse posttherapy. In a multivariable analysis, patients subsequently found to have renal abscess rather than pyelonephritis were more likely to have diabetes mellitus and to present with hypotension, acute renal impairment, and a peripheral leukocyte (white blood cell [WBC]) count $>20,000$ WBC/mL.²⁵⁹

CT is the preferred imaging modality for diagnosis; it also characterizes the abscess size and extent and may identify the potential source.¹²⁰ Ultrasonography identifies most perinephric abscesses but may not be able to distinguish an inflammatory lobar mass from a true renal abscess within the kidney.¹²¹

Management goals include prompt diagnosis, early effective antimicrobial therapy, and, in selected patients, abscess drainage for both therapeutic and diagnostic purposes. Culture of the abscess fluid identifies the specific infecting organisms and directs antimicrobial choice. If abscess fluid cannot be sampled, organisms isolated from blood or urine culture should be considered in the selection of antimicrobial therapy. Initial antimicrobial administration is usually intravenous. Once the patient's condition is stabilized, therapy can be continued with an appropriate oral antimicrobial agent that has good bioavailability. Small renal abscesses (up to 5 cm in diameter) may be managed medically, but larger abscesses usually require drainage.^{256,257,260-262} About 70% of renal abscesses resolve with medical therapy alone.²⁵⁷ Perinephric or mixed abscesses are larger, and drainage is usually required. The current initial approach for these larger abscesses is to attempt percutaneous drainage and, if this is not effective, to proceed to open drainage or nephrectomy. Resolution of the abscess is monitored by repeated imaging studies.

The duration of antimicrobial therapy is guided by the clinical course: It should be continued until the abscess has completely resolved or until there is only a residual, stable scar.

INFECTED RENAL CYSTS

The frequency of occurrence and risk factors for development of cyst infection are not well described. A French series reported that 8.4% of 389 patients with autosomal dominant polycystic kidney disease admitted to the nephrology department over a 10-year period had definite or likely cyst infection²⁶³; 24% of episodes were associated with bacteremia, and two patients had more than one episode. Infected cysts complicating polycystic kidney disease may be difficult to diagnose both clinically and microbiologically.²⁶⁴ Fever and abdominal pain or tenderness, often with bacteremia, are the usual clinical manifestations. However, the differential diagnosis includes pyelonephritis, infected kidney stones, perinephric abscess, cyst hemorrhage, and other intraabdominal pathology. Most patients with infected cysts also have peripheral leukocytosis and elevated CRP. *E. coli*, *K. pneumoniae*, *Enterococcus* species, and group B streptococci are the most common infecting organisms, but *Salmonella* spp., *P. aeruginosa*, *Clostridium* species, *Candida* species, and *Aspergillus* species have also been reported. In view of the wide spectrum of potential pathogens, the specific infecting organism should be confirmed by cyst aspiration whenever possible. Imaging studies are usually necessary to confirm the diagnosis and identify the implicated cyst. Ultrasonography and CT are not useful.²⁶³ MRI and WBC-labeled scans^{265,266} were successful in localizing the infected cyst in individual case reports. Positron emission tomography with fluorodeoxyglucose F-18 is reported to be most effective in localizing the infected cyst, but access to this technology is limited.^{264,267-269}

Once the potential infected cyst is identified, a cyst aspiration, if possible, can confirm the presence of infection and provide therapeutic drainage.²⁷⁰ Antimicrobial penetration into the cyst is presumed to be transepithelial. Effective concentrations of TMP/SMX, chloramphenicol, and fluoroquinolones are achieved in the cyst.²⁴³ Cyst and serum levels of levofloxacin are similar, whereas cyst levels reported with ciprofloxacin are only 40% of serum levels.²⁷¹ Therapeutic cyst antimicrobial levels are also achieved with penicillins, cephalosporins, aminoglycosides,²⁴³ and amphotericin B.²⁷² Prolonged antimicrobial therapy, for at least 4 weeks, is recommended, although clinical trials defining the optimal duration of therapy have not been reported. In the French case series described previously, the mean duration of antibiotic therapy was 5 weeks and the initial course of therapy was successful for 71% of cases.²⁶² Nephrectomy is occasionally necessary when cyst drainage and appropriate antimicrobial therapy are not successful in controlling the infection (see also Chapter 45).

EMPHYSEMATOUS CYSTITIS AND PYELONEPHRITIS

Emphysematous cystitis and pyelonephritis are acute necrotizing infections characterized by gas formation. Gas within the urinary collecting system can be seen after many interventional procedures, but emphysematous infection

is characterized by gas in the tissues.¹²¹ Gas is localized to the bladder wall and lumen in cystitis²⁷³ and in and around the kidney in pyelonephritis.²⁷⁴ Emphysematous pyelitis is a gas-forming infection restricted to the collecting system; the renal parenchyma is spared. Affected patients usually have diabetes with poor glucose control. Obstruction is another common predisposing factor for emphysematous pyelonephritis. *E. coli* and *K. pneumoniae* are most commonly isolated. High levels of urinary glucose serve as a substrate for these bacteria; large amounts of gas are generated through natural fermentation. In affected patients who are not diabetic, protein fermentation is a proposed source of gas formation.²⁷³ CT is considered the optimal imaging technique for confirming emphysematous infection and characterizing the extent of involvement.^{120,121}

A review of 135 cases of emphysematous cystitis identified over 50 years reported that the median patient age was 66 years (range, <1 to 90 years); 64% were women, and 67% had diabetes.²⁷³ Presenting symptoms ranged from pneumaturia alone through irritative lower-tract voiding symptoms to severe illness suggestive of acute abdominal disease or sepsis; 7% of cases were asymptomatic. *E. coli* was isolated from 58%, *K. pneumoniae* from 21%, and *Clostridium* species and *Enterobacter aerogenes* each from 7% of cases. The diagnosis was apparent on a plain radiograph of the abdomen in 84% of cases. Medical management including antimicrobial therapy, bladder drainage, and glycemic control was usually effective. Surgical intervention was required for only 10% of cases and included cystectomy, partial cystectomy, and surgical debridement when concomitant emphysematous pyelonephritis was present. The overall mortality rate was only 7%.

Emphysematous pyelonephritis is a more serious infection with a substantially higher mortality rate.²⁷⁵ In case series of patients with emphysematous pyelonephritis, 62% to 100% were diabetic and the majority were women.²⁷⁶⁻²⁷⁸ Serum glucose levels were often high in diabetic patients, and serum CRP was generally elevated. *E. coli* was isolated from 45% to 54% of episodes, and bacteremia occurred in 20% to 50% of patients in whom blood cultures were performed. Plain abdominal radiographs identified gas in the kidney in only 50% of cases. Ultrasonography of the kidneys usually showed some abnormalities but was less accurate than CT in identifying emphysematous infection. Management includes antimicrobial therapy together with percutaneous or open drainage of abscesses and correction of obstruction.^{274,275,277-279} Aggressive glucose control and supportive care are also necessary. The mortality rate ranges from 7% to 20% in patients with less severe manifestations to 70% in patients with a fulminant course characterized by necrosis, intravascular thrombosis, and microabscess formation.

Emergency nephrectomy was previously considered necessary for any patient presenting with emphysematous pyelonephritis. Currently, percutaneous drainage is the recommended initial approach. This is associated with similar or lower mortality rates than either emergency nephrectomy or medical management alone.^{275,279-282} In a retrospective case series describing a relatively small number of patients, the mortality rates were 50% with medical management alone, 25% with medical management and emergency nephrectomy, and 13.5% with medical management

combined with initial percutaneous drainage.²⁷⁶ Delayed elective nephrectomy may subsequently be required for some patients.

XANTHOGRANULOMATOUS PYELONEPHRITIS

Xanthogranulomatous pyelonephritis is an uncommon, severe, subacute, or chronic suppurative process characterized by destruction and replacement of the renal parenchyma by granulomatous tissue containing histocytes and foamy cells.^{283,284} The inflammatory process may extend into perinephric structures such as Gerota fascia, the posterior perirenal space, the psoas muscle, the diaphragm, and the spleen. Less than 1% to 8% of kidneys removed or subjected to biopsy for inflammatory conditions are reported to show evidence of xanthogranulomatous pyelonephritis.²⁸⁴ The pathogenesis of this process is unknown, but potential contributing factors include chronic UTI, abnormal lipid metabolism, lymphatic obstruction, impaired leukocyte function, and vascular occlusion. Urine culture results are positive in 62% to 89% of cases. Organisms isolated from renal tissue are *P. mirabilis* (38%), *E. coli* (33%), *Klebsiella/Enterobacter* species (8%), *P. aeruginosa* (8%), and *S. aureus* (10%).

In a single-center experience, 35 (85%) of 41 cases presenting from 1994 to 2005 were women.²⁸⁵ Xanthogranulomatous pyelonephritis was responsible for 19% of 214 nephrectomies performed at this facility during the review period. The most common presenting symptoms were fever, flank or abdominal pain, weight loss, lower urinary tract symptoms, and gross hematuria. All patients had renal calculi. Of 17 cases in which urine culture was obtained, *E. coli* was isolated in 35% and *P. mirabilis* in 18%; in 35% there was no growth. A review from Greece of 39 cases occurring between 1980 and 1999 reported a female-to-male ratio of 2:1.²⁸³ The presenting history included urolithiasis or renal colic, recurrent UTIs, and previous urologic procedures. All patients were symptomatic at presentation; symptoms included complaints of fever, flank or abdominal pain, chills, and malaise. Anorexia, weight loss, lower-tract symptoms, and gross hematuria were also reported. *E. coli* was isolated in 15 cases and *P. mirabilis* in 12. Plain radiographs or intravenous pyelography (IVP) showed a nonfunctioning kidney in 63% of cases, single or multiple calculi in 52%, staghorn calculus in 48%, calyceal deformity in 26%, and hydronephrosis in 23%. In 35 New Zealand patients presenting from 2001 to 2013,²⁸⁶ 91% were female, staghorn calculi occurred in 51%, and obstructing ureteric calculi in 23%. The most common organisms isolated were *E. coli* (46%) and *P. mirabilis* (20%). A more recent review of 40 patients from India reveals similar patterns: Most patients were women, and 90% had nephrolithiasis. All patients had unilateral disease, and all underwent nephrectomy.²⁸⁷

CT is considered the optimal imaging modality; it identifies the abnormality in 74% to 90% of cases.^{120,121,288} Because of the current widespread access to CT, the diagnosis is now usually made preoperatively. Characteristic findings include an enlarged kidney, frequently with replacement of renal parenchyma and multiple fluid-filled cavities, together with urolithiasis. Ultrasonography identifies nonspecific abnormalities including renal enlargement with relative preservation of renal contour and multiple hypoechoic

round masses.¹²¹ MRI shows abnormalities similar to those observed with CT.²⁸⁸ The differential diagnosis includes malignancy and tuberculosis. The usual management is nephrectomy; antimicrobial therapy has only a secondary role.²⁸⁴ If the diagnosis is made early, when there is only focal renal involvement, partial nephrectomy may be curative.

PYOCYSTIS

Pyocystis, also called “vesicle empyema,” is an infection characterized by a purulent fluid collection in the bladder of a patient with a nonfunctioning bladder. In effect, the bladder becomes an undrained abscess. This is a rare complication diagnosed in patients with anuric renal failure or surgically bypassed bladders. The clinical presentation includes suprapubic pain or distention, abdominal pain, foul-smelling urethral discharge, and fever or sepsis.^{289,290} Organisms isolated have included *E. coli*, *P. mirabilis*, *P. aeruginosa*, *Serratia marcescens*, *Streptococcus* species, *Enterococcus* species, and *Candida* species. Mixed cultures are common. It is not known whether anaerobic organisms may also contribute.

When the condition is suspected, a specimen of bladder fluid should be obtained for diagnosis and culture. The laboratory should be requested to identify all organisms isolated so that the specimen is processed as abscess fluid rather than urine. The treatment approach involves systemic antimicrobial agents and urethral catheterization for bladder drainage. Antimicrobial therapy is directed by the specific organisms isolated. Bladder irrigation with either saline or an antibiotic solution is sometimes recommended, but whether irrigation provides an additional therapeutic benefit is not clear.²⁹¹ Surgical intervention to achieve adequate drainage is necessary, rarely, for recalcitrant cases.

UNCOMMON ORGANISMS

GENITOURINARY TUBERCULOSIS

Genitourinary tuberculosis is diagnosed in 1.1% to 1.5% of all tuberculosis cases and 5% to 6% of cases of extrapulmonary tuberculosis.²⁹²⁻²⁹⁴ This infection is usually a consequence of local reactivation following hematogenous dissemination of *Mycobacterium tuberculosis* to the renal cortex during primary pulmonary infection. The renal cortex is also frequently involved with miliary disease, with multiple granulomas usually present. The high oxygen tension of the renal cortex is favorable for renal localization. Men are infected twice as often as women. The latent period, from the time of initial pulmonary infection to diagnosis of clinical urogenital tuberculosis, is 22 years on average with a range of 1 to 46 years.²⁹⁵ Reactivation of tuberculosis usually occurs in only one kidney; thus disease is characteristically unilateral. Contiguous involvement of the collecting system leads to *M. tuberculosis* bacilluria with subsequent ureteric and bladder infection. Prostate and epididymis infection may result directly from hematogenous dissemination rather than contiguous spread. The kidneys are affected in 60% to 100% of cases, the ureters in 19% to 41%, the bladder in 15% to 20%, and the prostate or epididymis in 20% to 50% of men.²⁹⁵⁻²⁹⁷

Substantial morbidity is attributed to urogenital tuberculosis. Severe calyceal clubbing and dilation of the renal pelvis and ureters leading to total destruction of the kidney and autonephrectomy occur in 23% to 33% of cases, and renal failure occurs in 1% to 10%.²⁹⁵ A summary of cases reported worldwide found that 27% of patients had a nonfunctioning unilateral kidney at presentation, but this proportion varied from 8% to 72% in different countries, presumably reflecting the timeliness of diagnosis.²⁹⁸

Presenting genitourinary symptoms are often vague or nonspecific; they include back or flank pain, dysuria, and urinary frequency and, in women, infertility. Up to 50% of patients have no localizing genitourinary symptoms. About 25% to 33% of patients have systemic complaints, usually pulmonary symptoms, fever, and weight loss.^{295,299} The subacute presentation and results of the initial evaluation can mimic those of xanthogranulomatous pyelonephritis. In men, additional features suggestive of tuberculosis include an enlarged, hard, and nontender epididymis; thickened or beaded vas deferens; indurated or nodular prostate; and nontender testicular mass.²⁹³ Tuberculous granulomatous prostatitis may manifest as a nodular prostate with an elevation of prostate-specific antigen that is clinically indistinguishable from that in prostate carcinoma. Most patients with renal tuberculosis have evidence of concomitant extragenital disease. The most common site is the lungs, but the pulmonary disease is usually inactive. The chest radiograph is abnormal in 67% to 75% of patients, and the tuberculin skin test positive in 60% to 90%.

Urinalysis findings are abnormal in >90% of patients with genitourinary tuberculosis.²⁹² Sterile pyuria with hematuria is most common, but sterile pyuria alone and gross or microscopic hematuria may also occur.^{292,296,297} For patients with HIV infection, an algorithm that incorporates negative results of routine urine culture and the presence of pyuria, albuminuria, or hematuria has good predictive value for detecting genitourinary tuberculosis.^{300,301} As many as half of HIV-positive patients, however, have concomitant bacteriuria with other organisms at presentation, and this finding may obscure the initial assessment of the urinalysis.

Intravenous pyelography was traditionally the standard imaging approach, but CT is now preferred.²⁹⁵ Imaging studies identify unilateral disease in 75% of cases of renal tuberculosis (Fig. 38.1). The characteristic early finding is erosion of the renal calyx, which subsequently progresses to papillary necrosis, hydronephrosis, renal parenchymal cavitation, and dilated calyces. Ureteric tuberculosis is characterized by a thickened ureteric wall and strictures. Lesions are most common in the distal third of the ureter. Bladder tuberculosis may manifest as reduced bladder volume with wall thickening, ulceration, and filling defects resulting from granulomatous involvement. In advanced disease, scarring results in permanent loss of volume and a residual small, irregular, calcified bladder. The most common finding on CT is renal calcification, present in 50% of cases.³⁰² Other characteristic findings are hydrocalyx secondary to infundibular stenosis and cavity formation. In advanced disease, cortical loss, uroepithelial thickening, and dystrophic calcification are present.

The diagnosis is confirmed by growth of *M. tuberculosis* in urine or tissue culture. Appropriate specimens for

mycobacterial culture should be obtained whenever this diagnosis is considered. Three sequential early morning urine specimens for mycobacterial culture are recommended. Urine cultures have positive results in 75% to 90% of affected patients. A positive acid-fast bacillus smear of urine usually signifies *M. tuberculosis*, but culture confirmation is essential to rule out colonization with nonpathogenic mycobacteria and for susceptibility testing of the infecting strain.^{303,304} If a suggestive renal abnormality is present but urine culture results are negative, tissue biopsy may be necessary to confirm the diagnosis. PCR nucleic acid antigen testing of urine specimens has been reported to be more sensitive than culture and, if available, provides a more rapid diagnosis.^{294,301} However, culture is still necessary to isolate the organism for susceptibility testing.

Antimycobacterial treatment should be provided under the supervision of a physician with expertise in tuberculosis management. The treatment of genitourinary tuberculosis is similar to that of extrapulmonary tuberculosis at other sites. The initial regimen consists of four drugs (isoniazid, rifampin, pyrazinamide, and ethambutol) for 2 months, followed by two drugs (isoniazid and rifampin) for 4 months if the isolate is susceptible to first-line therapy. Follow-up IVP every 6 months has been recommended to identify ureteral scarring or obstruction, which may develop or progress during therapy as part of the healing process. Corticosteroid therapy does not prevent this complication.²⁹² Ureteral reimplantation, endoscopic balloon dilation, or implantation of ureteral stents may be necessary if progressive obstruction develops.³⁰⁵ Nephrectomy is rarely required but may be indicated for intractable pain, untreatable infection proximal to a stricture, uncontrollable hematuria or hypertension, or drug resistance. Bladder augmentation surgery may be required if the bladder is scarred and contracted following treated tuberculosis infection.

BACILLUS CALMETTE–GUÉRIN INFECTION

Intravesical vaccine instillation of bacillus Calmette–Guérin (BCG) is considered first-line treatment for superficial bladder tumors and carcinoma in situ. Treatment with this biologic therapy is complicated by systemic or local BCG infection in <5% of cases.^{306–308} BCG bladder instillation is followed by local irritative symptoms such as dysuria in 80% of cases, but these symptoms do not usually persist beyond 48 hours.³⁰⁹ If symptoms do persist, isoniazid for 14 days is recommended. When symptoms continue despite isoniazid therapy, a full course of antituberculous therapy is recommended³¹⁰ and may need to be prolonged.³¹¹ Less common genitourinary infections that accompany BCG instillation include prostatitis in 1% to 3% of patients, epididymitis in 0.2%, and, in rare cases, testicular abscesses, bladder ulcers, local skin infections, or renal infection. Reflux may be a risk factor for the rare complication of BCG pyelonephritis.³¹² Localized genitourinary infection tends to have a delayed onset and is usually not apparent clinically until more than 3 months after BCG treatment.^{305,306} BCG may not be isolated from urine or tissue cultures.³⁰⁶ Tissue biopsy usually shows necrotizing granulomatous inflammation, and this finding in the context of prior BCG therapy is sufficient for diagnosis even if subsequent culture results are negative.³⁰⁸

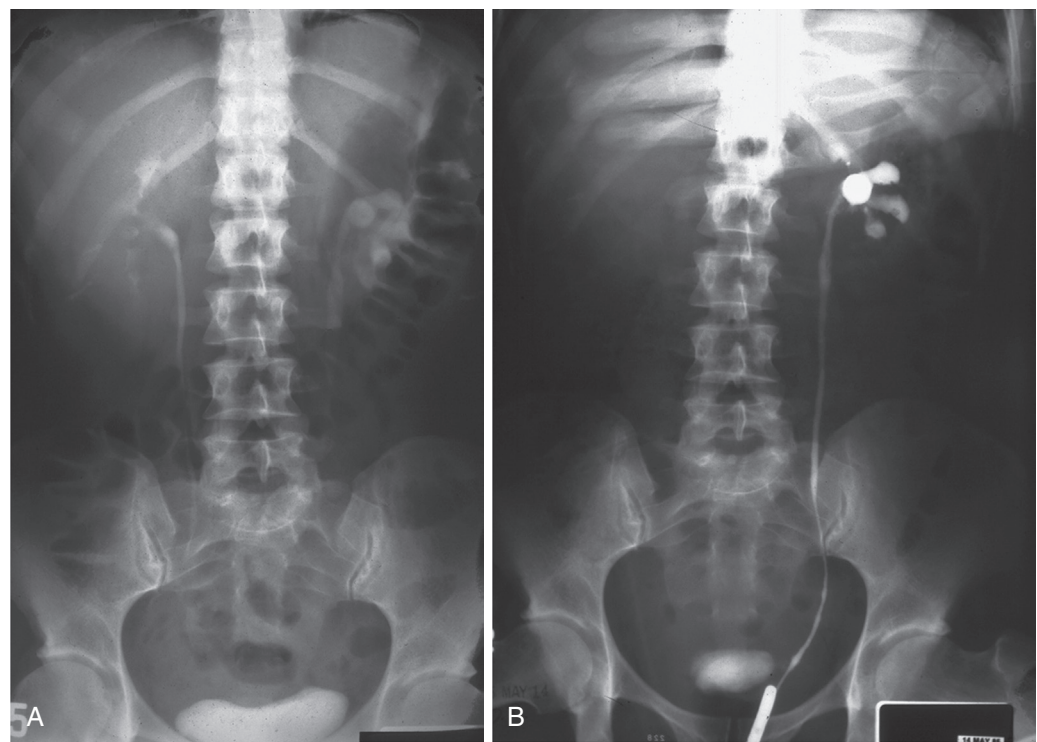


Fig. 38.1 Renal tuberculosis. (A) Intravenous pyelogram, showing unilateral hydronephrosis and calyceal distention. (B) Retrograde pyelogram in the same patient, showing distal ureteric narrowing.

FUNGAL URINARY TRACT INFECTION

Fungal UTI is usually caused by *Candida* species. These organisms have an extensive array of virulence factors that may facilitate successful colonization and invasion of the urinary tract.³¹³ Infection may be either antegrade, following candidemia, or retrograde via the urethra.³¹³ Clinical manifestations are asymptomatic candiduria, cystitis, and pyelonephritis. Bladder or renal fungus balls and systemic fungemia are rare complications.³¹⁴

Candida albicans is isolated from more than 50% of episodes, followed by *Candida glabrata*, *Candida tropicalis*, and *Candida parapsilosis*.^{315,316} Candiduria is usually identified in patients who are seriously ill with multiple comorbid conditions, and most infections are asymptomatic.³¹⁴ The most important risk factors for candiduria are the presence of an indwelling catheter or other indwelling urologic device, diabetes mellitus, and exposure to broad-spectrum antimicrobial agents.³¹⁶ Treatment of asymptomatic candiduria is not beneficial and is currently recommended only for selected patients with high-risk neutropenia (<100 WBC/mm³) or before a traumatic urologic procedure.^{317,318} Indwelling devices should be removed as soon as possible to facilitate resolution of candiduria. Imaging studies to exclude fungus balls should be considered in patients who have recurrent symptomatic infection or urinary obstruction.³¹⁹ When fungus balls are present, surgical removal is required for cure of the infection.

Fluconazole is the treatment of choice for *Candida* UTI because it is excreted in the urine in active form and high urinary levels are achieved.³¹⁷ A 1- to 2-week course of fluconazole, 200 to 400 mg daily, is recommended (Table 38.7). Amphotericin B deoxycholate, 0.3 to 0.6 mg/kg daily, is the alternative treatment recommended

Table 38.7 Treatment of Candiduria

	Cystitis	Pyelonephritis
Fluconazole (if isolate is susceptible)	200-400 mg daily, 7-14 days	200-400 mg daily, 14 days
Amphotericin B deoxycholate	0.3-0.6 mg/kg daily, 1-7 days	0.5-0.7 mg/kg daily, 14 days
Amphotericin B bladder irrigation	5-50 mg/L continuous irrigation for 2-7 days	Not indicated
Flucytosine	25 mg/kg qid, 7-10 days	25 mg/kg qid, 14 days ^a

^aMay be used in combination with amphotericin B. qid, Four times a day. Adapted from Pappas PG, Kauffman CA, Andes DR, et al. Clinical practice guideline for the management of candidiasis: 2016 update by the Infectious Diseases Society of America. *Clin Infect*

for fluconazole-resistant strains including most *C. glabrata* organisms, as well as in patients who are allergic to fluconazole or in whom treatment fails despite optimal fluconazole therapy and urologic management. Recommended treatment duration with amphotericin B is 1 to 7 days for cystitis and 1 to 2 weeks for pyelonephritis. Amphotericin B may also be effective for treatment of fungal cystitis when administered as a bladder washout, but this approach is not generally recommended because it requires several days of urethral catheterization and the optimal dose, frequency,

and duration are not well established.³²⁰ With the lipid formulations of amphotericin B, concentrations of active drug achieved in renal tissue and urine are low, so these formulations are not recommended.

The antifungal 5-flucytosine is excreted in the urine and indicated for treatment of *Candida* UTI as a single agent or in combination with amphotericin B for renal infection.³¹⁷ Resistance to 5-flucytosine develops rapidly when this drug is used as a single agent, and the frequent adverse effects of bone marrow suppression and enterocolitis also limit its use, particularly in patients with renal failure. Echinocandins such as caspofungin, micafungin, and anidulafungin and other azoles, such as itraconazole, voriconazole, and posaconazole, which are not excreted into the urine, are not recommended for treatment of UTI.³¹⁷ There are, however, case reports of successful treatment of *C. glabrata* UTI with caspofungin,³²¹ and a case series of selected patients treated with micafungin reported a success rate of 75%.³²²

VIRAL INFECTIONS

Viral UTIs are uncommon in adults and occur largely in immunocompromised patients.³²³⁻³²⁶ Clinical manifestations of viral infection generally follow reactivation of latent infection in immunocompromised patients, although de novo infection may occur. The usual clinical manifestation is hemorrhagic cystitis, but nephropathy has also been described.^{323,327-329} The most common viruses are adenovirus, parvovirus B19, and cytomegalovirus (CMV). More than one viral infection may coexist. Most adult cases occur in recipients of hematopoietic stem cell transplants, particularly those with severe graft-versus-host disease, and in renal transplant recipients. Infection has also been reported in other immunosuppressed patients, such as HIV-infected patients with low CD4⁺ cell counts.^{325,330} Adenovirus or CMV infection is diagnosed by viral culture or PCR of the urine. Management includes minimization of immunosuppressive therapy if possible. For HIV-infected patients, antiretroviral therapy to increase the CD4⁺ cell count should be initiated. CMV infection typically responds to treatment with ganciclovir or foscarnet.³²³ Treatment of adenovirus infection uses cidofovir, although some efficacy has been reported with vidarabine.³²³ There are also case reports of successful treatment of adenovirus with ganciclovir and ribavirin.^{323,331}

PARASITIC INFESTATIONS OF THE URINARY TRACT

The most common and important parasitic infestation of the urinary tract is with *Schistosoma haematobium*.³³²⁻³³⁴ This parasite is acquired after exposure to contaminated water. *Schistosoma* larvae penetrate the intact skin and migrate in the blood to the liver, where they transform into young worms (schistosomulae) that mature in 4 to 6 weeks and then further migrate to the perivesical venules. The life span of the adult worm in the venule is usually 3 to 5 years but can be longer. Most eggs produced by the adult worms enter the bladder lumen and are removed in the urine. However, some eggs are retained locally in the bladder wall, where they incite an eosinophilic inflammatory and granulomatous immune response that causes progressive fibrosis. The functional abnormality early in the disease

is obstruction of the bladder neck. Late complications include recurrent bacterial UTI, bladder or ureteric stone formation, renal functional abnormalities, and, ultimately, kidney failure.^{333,335} *Schistosoma* infestation is a risk factor for squamous cell carcinoma of the bladder,³³⁶ with relative risk ranging from 1.8 to 23.5; it is highest in the 30- to 50-year age group.³³⁷

Prevalence of infection in endemic areas is high. Surveys from rural Zimbabwe revealed that 60% of women younger than 20 years of age and 29% of those aged 45 to 49 years have eggs in the urine; HIV-infected women older than 35 years had a significantly higher prevalence of infestation.³³⁸ Prevalence in a recent study of 245 preschool-aged children in Angola was 30%.³³⁹ Travelers to endemic areas may acquire infection with only minimal exposure to contaminated water³⁴⁰⁻³⁴² (see also [Chapters 75–77](#) for discussion of schistosomiasis in Latin America, Africa, and the Near and Middle East).

Acute genitourinary symptoms occur in up to 50% of cases following acquisition of infection and include hematuria, which is often terminal, together with dysuria and urinary frequency. Microhematuria is reported in 41% to 100% and gross hematuria in 0% to 97% of patients with chronic schistosomal infestation. Radiologic abnormalities are present in the upper urinary tract in 2% to 62% of chronic cases.³³³ Ultrasonography of the urinary tract demonstrates thickening of the bladder wall, granulomatous changes, hydronephrosis, and, on occasion, bladder or ureteric calcification. The diagnosis is established by identification of parasite eggs in the urine or biopsy specimens or by serologic findings.³³⁴ Urine specimens collected for identification of eggs should be obtained on consecutive days between 1100 and 1300 hours because egg passage is maximal at this time. Sedimentation or filtration of the urine before examination increases the sensitivity of egg detection.^{333,343}

Treatment with one dose of praziquantel, 40 mg/kg body weight, will cure 80% of cases.^{334,344} Follow-up urine specimens for parasite examination are recommended 3 months after treatment to identify patients in whom treatment has failed and must be repeated. Bladder wall thickening and hydronephrosis may be reversed in the majority of affected patients if treatment is given early in the course of infestation.³⁴⁵ However, when chronic disease is established and fibrotic lesions are present, changes may not be reversible and corrective surgery or management of end-stage renal disease is required.³⁴³

The protozoal parasite *T. vaginalis* is commonly transmitted sexually and is occasionally identified on microscopy with routine urinalysis. In women, the parasite may originate from contamination of the urine by vaginal secretions, but the organism is a well-described cause of urethritis for both men and women. Whenever *T. vaginalis* is identified, treatment of the patient and his or her sexual partners is indicated, regardless of symptoms. The recommended treatment is a metronidazole 500 mg twice daily for 7 days,³⁴⁶ or tinidazole, 2 g.³⁴⁷

Echinococcus granulosus infestation occasionally involves the kidneys.³²⁹ Renal cysts are reported in 2% to 3% of cases of hydatid disease.^{348,349} The diagnosis is usually made after an incidental finding of a cyst in the kidneys, ureters, bladder, or testes by imaging or in the investigation of nonspecific symptoms. On occasion, flank pain or a mass is present.

Hydatid cysts are not excreted in the urine. Treatment consists of surgical cyst removal or marsupialization; nephrectomy is occasionally necessary.³⁵⁰ Perioperative albendazole therapy is also usually recommended for patients with hydatid disease. A less common helminthic infestation is *Wuchereria bancrofti* (filariasis), which may cause lymphatic obstruction and rupture into the urinary collecting system, producing chyluria.³³²

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QUESTIONS

1. A 45-year-old woman presents for routine assessment 6 months after renal transplant. She has no general or specific clinical complaints. A routine urine culture grows *Escherichia coli* $>10^5$ resistant to fluoroquinolones. The most appropriate management of the urine culture result for this woman is:
 - a. Treat with a 7-day course of ertapenem
 - b. Repeat the urine culture
 - c. No interventions are necessary
 - d. Advise the patient to take cranberry capsules
 - e. Request an ultrasound examination

Answer: c

Rationale: This is asymptomatic bacteriuria. There is no indication for treatment of asymptomatic bacteriuria in otherwise well renal transplant patients who are 6 months post surgery. Treatment of asymptomatic bacteriuria does not decrease the frequency of symptomatic urinary tract infection or improve outcomes related to graft function, hospitalization, or other morbidity. Antimicrobial treatment of asymptomatic bacteriuria will encourage further emergence of more resistant bacteria, which could compromise future management of symptomatic infection, should it occur.

2. A 24-year-old sexually active university student presents with acute left-sided costovertebral angle pain associated with nausea and vomiting and is found to have a temperature of 39.2°C. A diagnosis of acute pyelonephritis is made. The patient has no history of recent travel or antimicrobial therapy within the previous 6 months. The appropriate initial management is:
 - a. Admit to hospital for treatment with ertapenem.
 - b. Provide a 7-day course of oral levofloxacin 750 mg daily.
 - c. Obtain urine culture and manage symptomatically with antiemetics pending results of urine culture and bacterial susceptibilities.
 - d. Give a single IV dose of ceftriaxone or gentamicin, followed by an oral course of ciprofloxacin 500 mg bid for 7 days.
 - e. Treat with gentamicin 180 mg once a day for 7 days.

Answer: d

Rationale: The current recommended treatment of acute pyelonephritis is immediate empiric institution of antimicrobial therapy. The recommended therapy is a fluoroquinolone, either ciprofloxacin 500 mg bid for 7 days or levofloxacin 750 mg od for 7 days, with or without an initial single dose of parenteral ceftriaxone or aminoglycoside. The presence of vomiting in this case would support the use of an initial parenteral antimicrobial dose. With the increasing occurrence of

extended-spectrum β -lactamase (ESBL)-producing *E. coli* in the community, which are also usually resistant to fluoroquinolones, initiation of ertapenem pending urine culture results may be considered when the patient is severely ill and considered at high risk for infection with an ESBL *E. coli*. High-risk patients would be those with recent travel to areas with a high prevalence of ESBL, who have received recent antimicrobial therapy, or who reside in a community where the local prevalence of ESBL is known to be high. In all cases, the urine culture results should be reviewed once available, usually at 48 to 72 hours, to ensure the empiric therapy initiated is effective for the organism isolated. If the organism isolated is resistant to the empiric therapy, the patient should be further treated with a full course of antimicrobial therapy to which the organism is susceptible, irrespective of whether there has been clinical improvement with the initial empiric regimen.

3. A 62-year-old gentleman who was born and raised in the Philippines presents with vague and nonspecific symptoms of discomfort in his left flank. Urine culture is negative, but urinalysis is positive for pyuria and microscopic hematuria. An infused CT scan reveals a stricture of the left ureter and erosion of the left renal calyx. The appropriate microbiologic specimen to confirm a diagnosis of genitourinary tuberculosis in this gentleman is:
 - a. Urine for culture and sensitivity
 - b. Three consecutive early morning urine specimens sent for culture for *Mycobacterium tuberculosis*
 - c. Urine collected on three consecutive mornings for smear for acid-fast bacilli
 - d. Urine PCR for *Mycobacterium tuberculosis*
 - e. Urine specimen collected for *Mycobacterium tuberculosis*.

Answer: b

Rationale: This gentleman comes from a highly endemic area for tuberculosis, and the nonspecific presentation together with the CT findings and urinalysis abnormalities would be consistent with genitourinary tuberculosis. Microbiologic documentation is required to confirm the diagnosis before initiating treatment, and a culture method is preferred so that antimicrobial susceptibility testing can be performed to optimize antimycobacterial therapy. The preferred diagnostic test is three consecutive early morning urine specimens forwarded to the laboratory with a request for acid-fast bacilli (AFB) culture. Urine for AFB staining is relatively unhelpful because atypical mycobacteria may be present as colonizers but lead to a positive AFB result. Urine PCR techniques are becoming more available but do not provide information with respect to the antimicrobial susceptibility of the organism.